US012398684B2

(12) United States Patent
Kamat et al.

(10) **Patent No.: US 12,398,684 B2**
(45) **Date of Patent: *Aug. 26, 2025**

(54) **TURBOMACHINERY ENGINES WITH HIGH-SPEED LOW-PRESSURE TURBINES**

(71) Applicant: **General Electric Company**, Cincinnati, OH (US)

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(73) Assignee: **General Electric Company**, Cincinnati, OH (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/745,410**

(22) Filed: **Jun. 17, 2024**

(65) **Prior Publication Data**

US 2024/0337228 A1 Oct. 10, 2024

Related U.S. Application Data

(63) Continuation-in-part of application No. 18/318,604, filed on May 16, 2023, now Pat. No. 12,012,901.

(30) **Foreign Application Priority Data**

Feb. 17, 2023 (IN) 202311010789

(51) **Int. Cl.**

F02K 3/06	(2006.01)
F01D 5/14	(2006.01)
F02C 3/06	(2006.01)

(52) **U.S. Cl.**

CPC ***F02K 3/06*** (2013.01); ***F01D 5/143*** (2013.01); ***F02C 3/06*** (2013.01); *F05D 2200/14* (2013.01)

(58) **Field of Classification Search**

CPC . F02K 3/06; F02K 3/077; F01D 5/143; F01D 1/26; F02C 7/36; F02C 3/06

See application file for complete search history.

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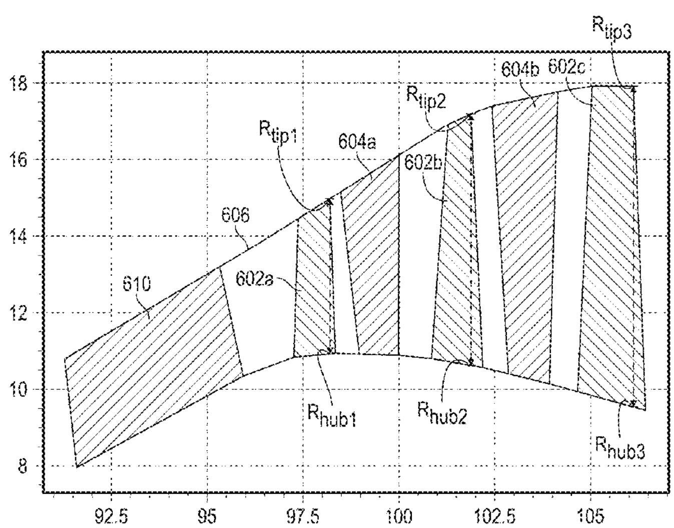
Primary Examiner — Topaz L. Elliott

(74) *Attorney, Agent, or Firm* — Klarquist Sparkman, LLP

(57) **ABSTRACT**

A turbomachinery engine includes a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine includes four rotating stages. The low-pressure turbine includes an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine. In some instances, the area ratio is within a range of 2.0-5.1. Additionally (or alternatively) the low-pressure turbine includes an area-EGT ratio within a range of 1.05-1.6.

7 Claims, 21 Drawing Sheets



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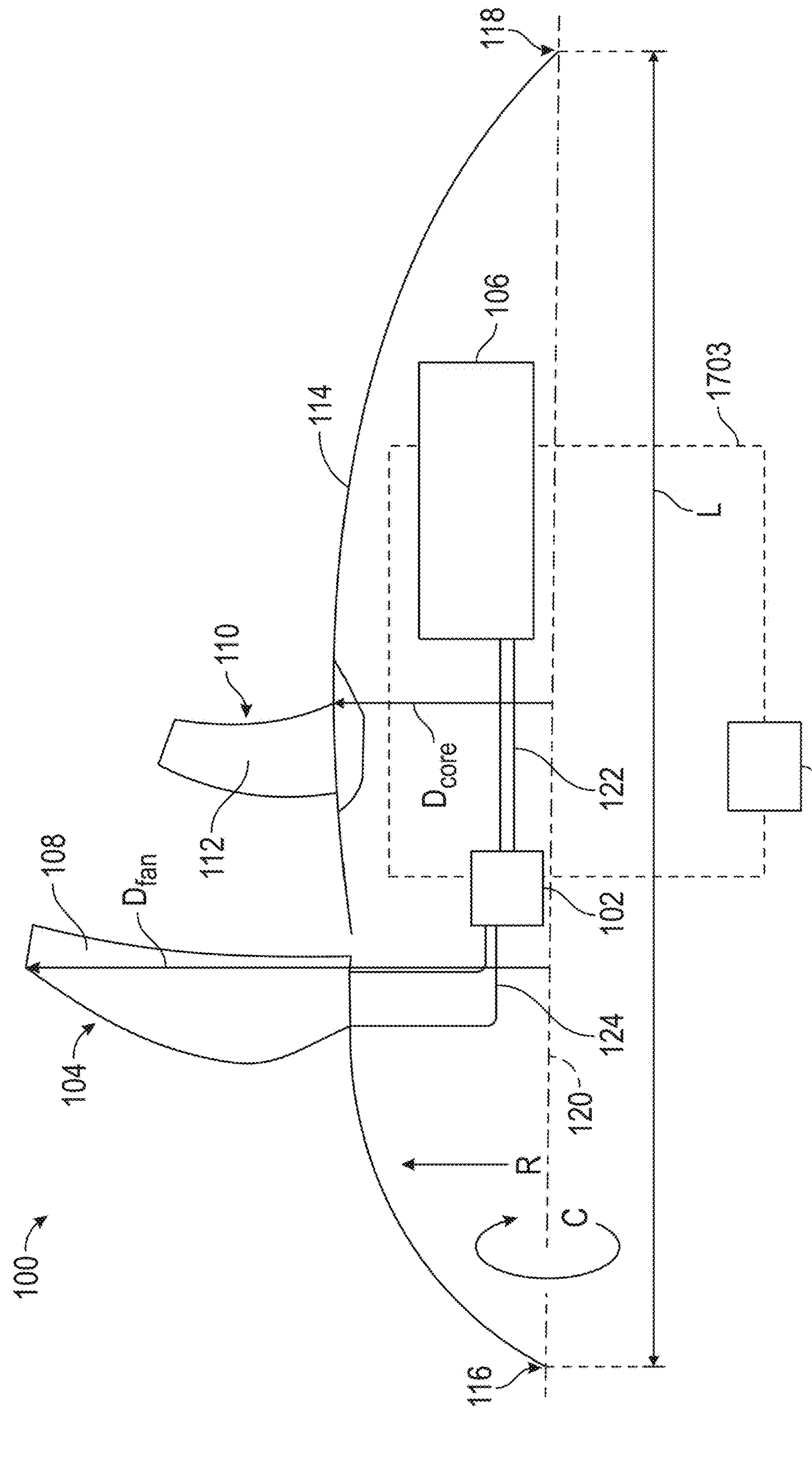
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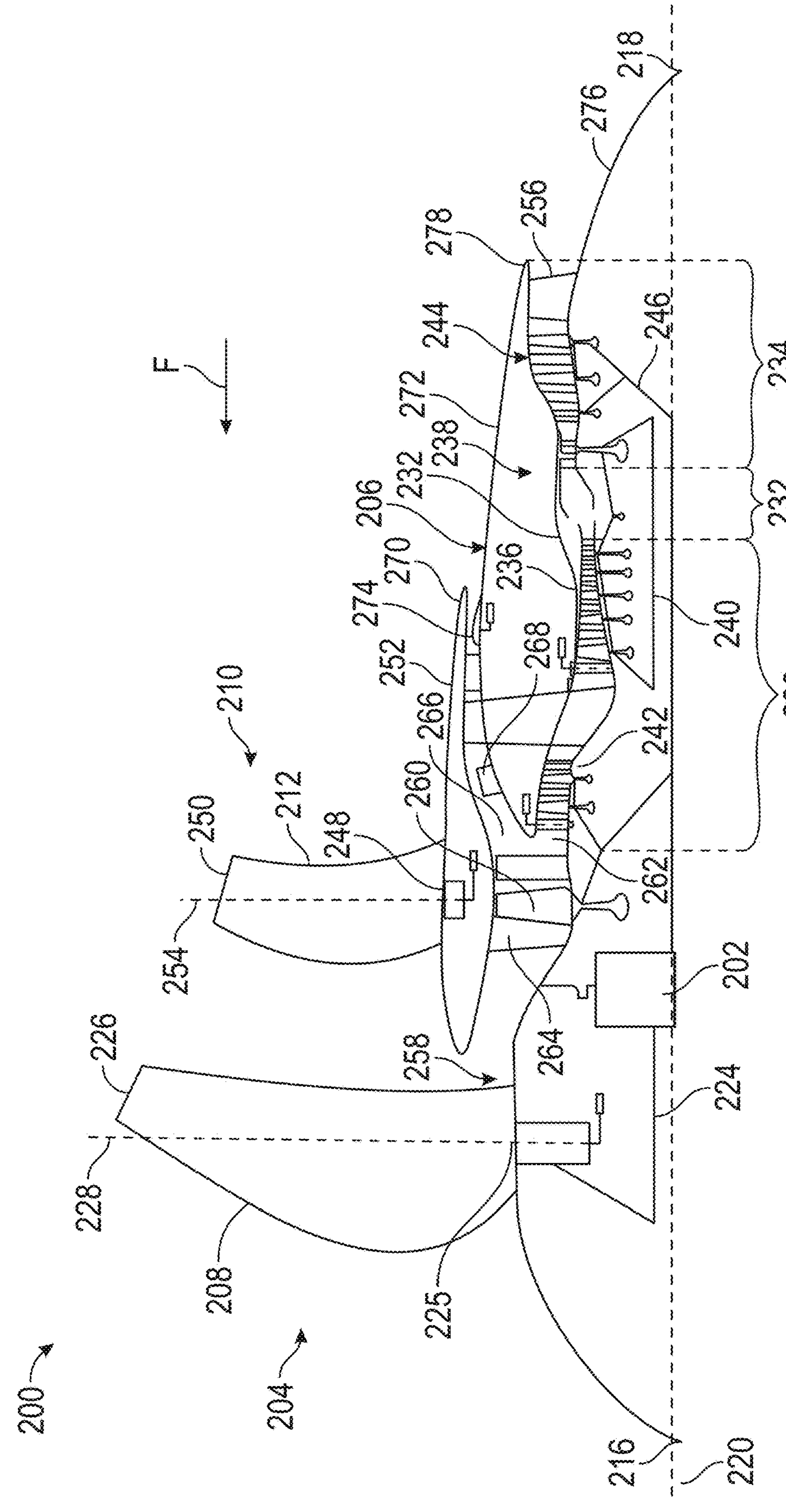
FIG. 1



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226
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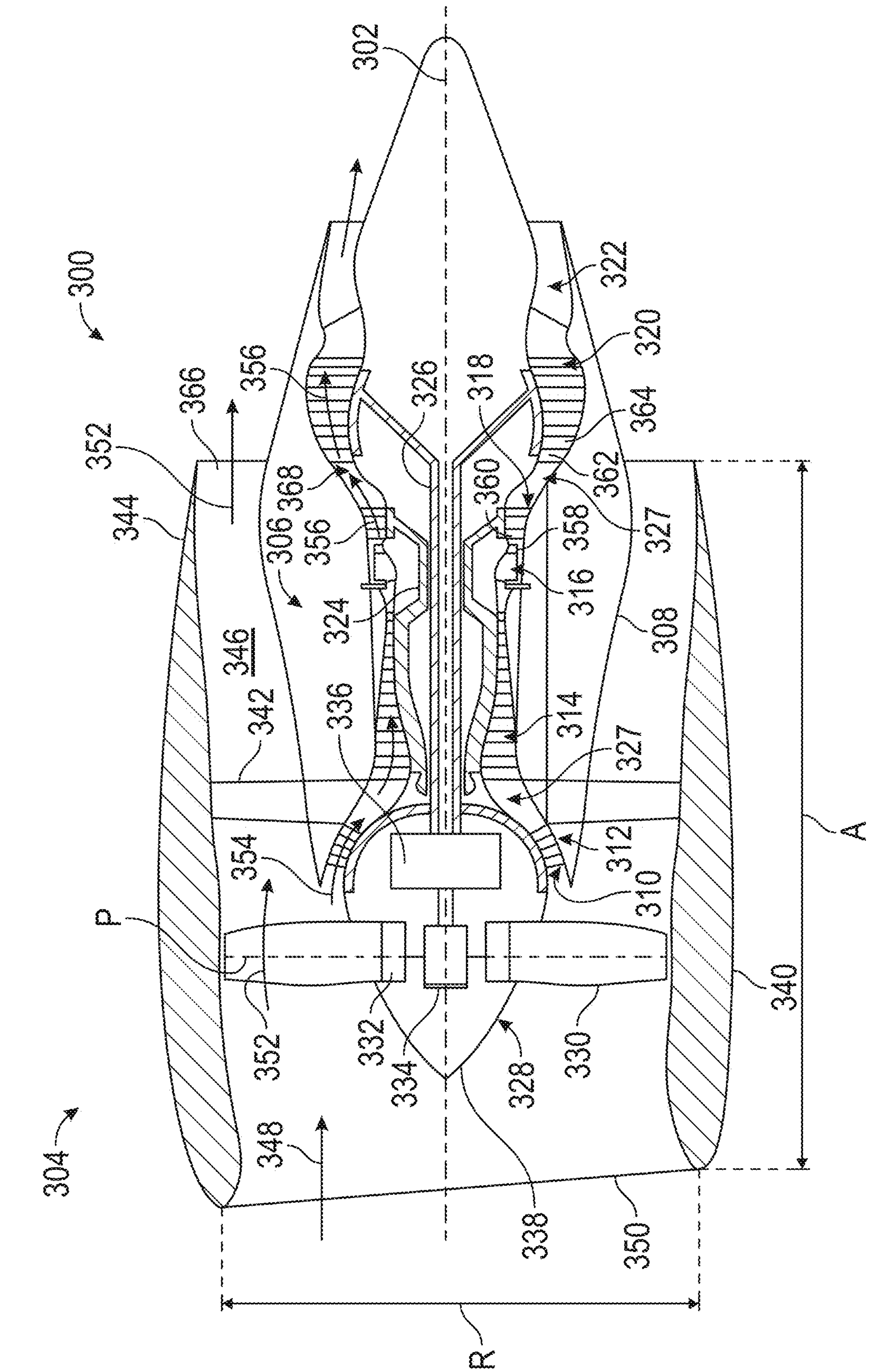
FIG. 2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338
340
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360
362
364
366
368
P
A
R

FIG. 3

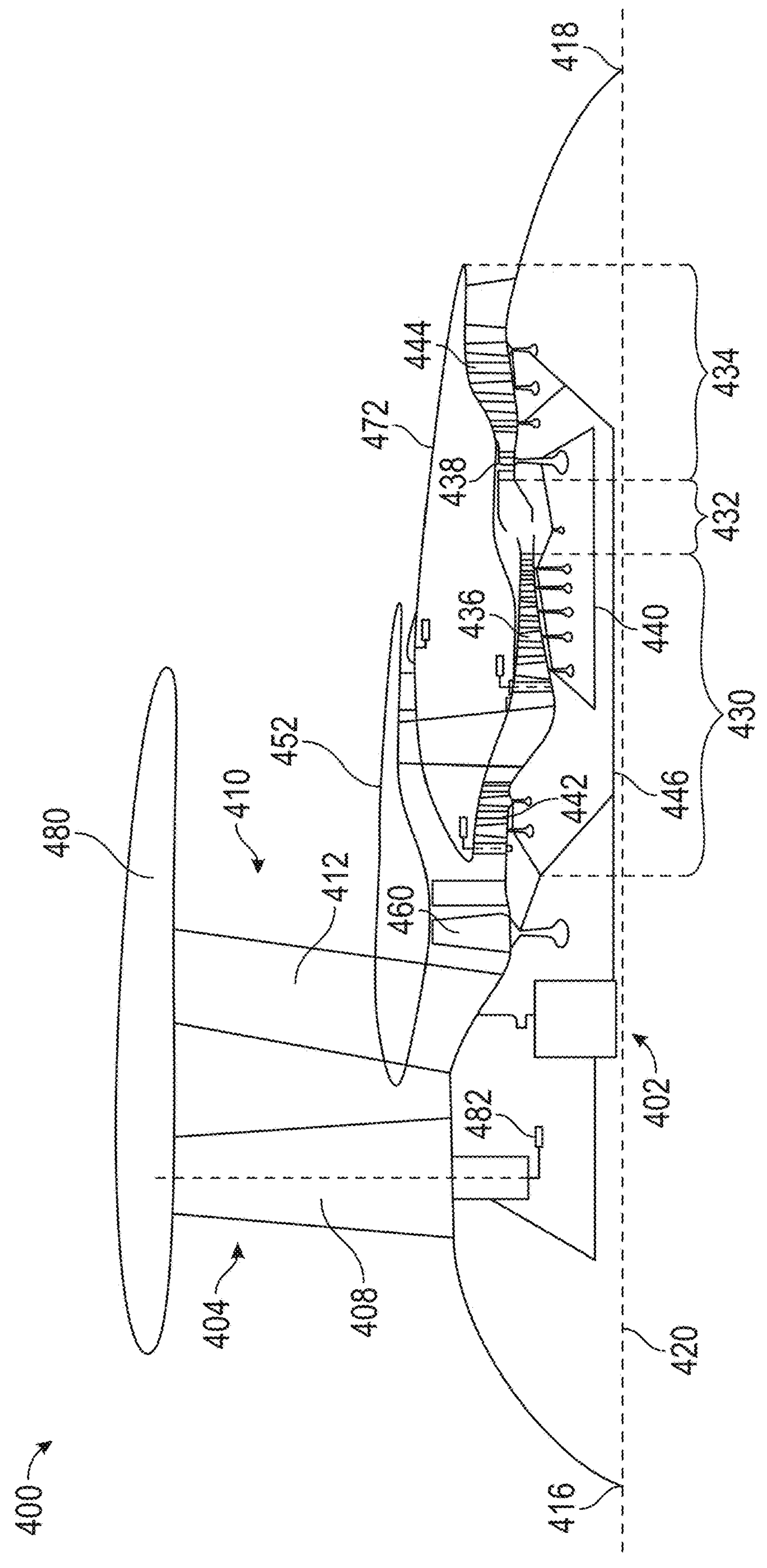
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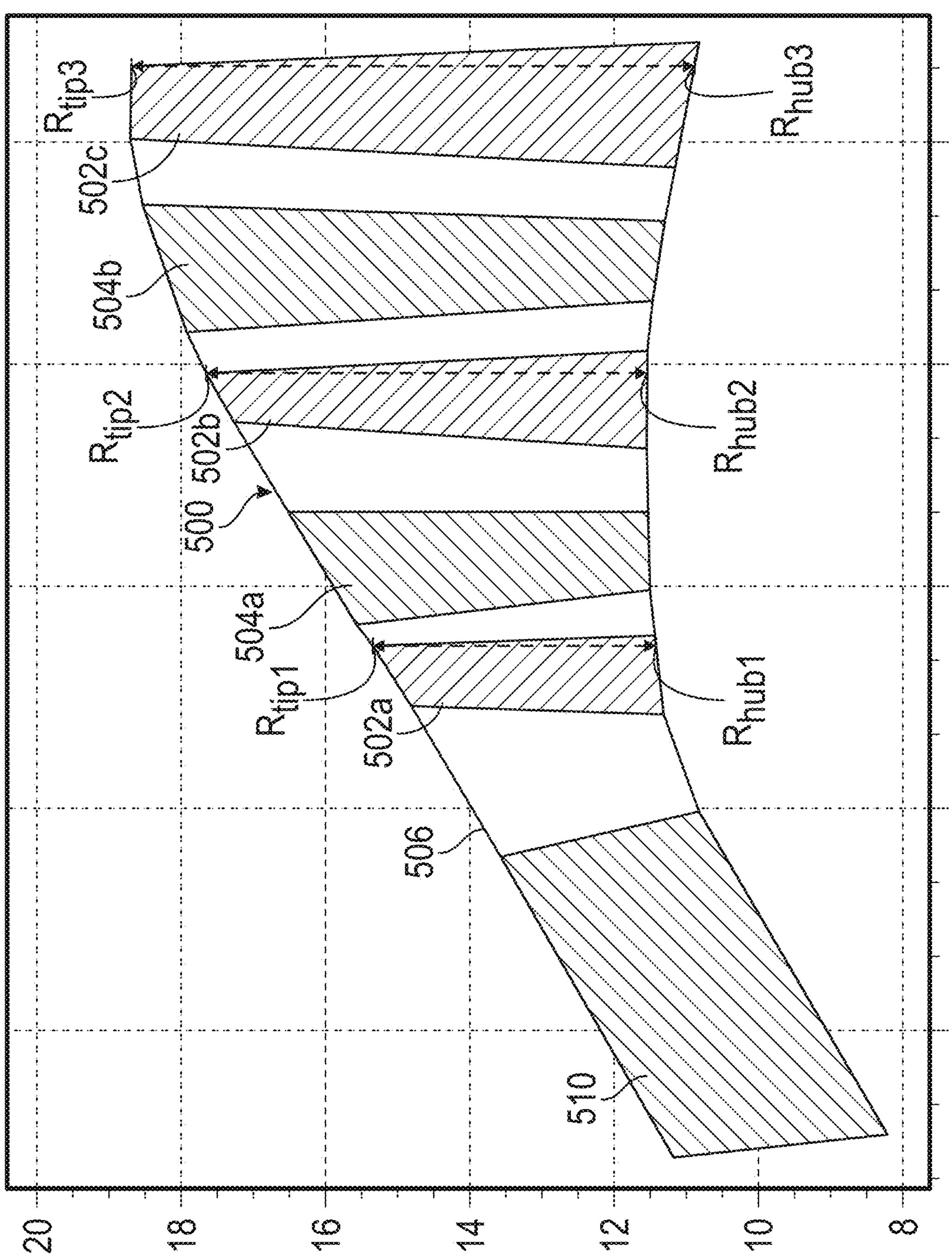
FIG. 4



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R_{hub3}
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504b
R_{tip2}
502b
R_{hub2}
500
504a
R_{tip1}
502a
R_{hub1}
506
510
93
96
99
102
105
20
18
16
14
12
10
8

FIG. 5A

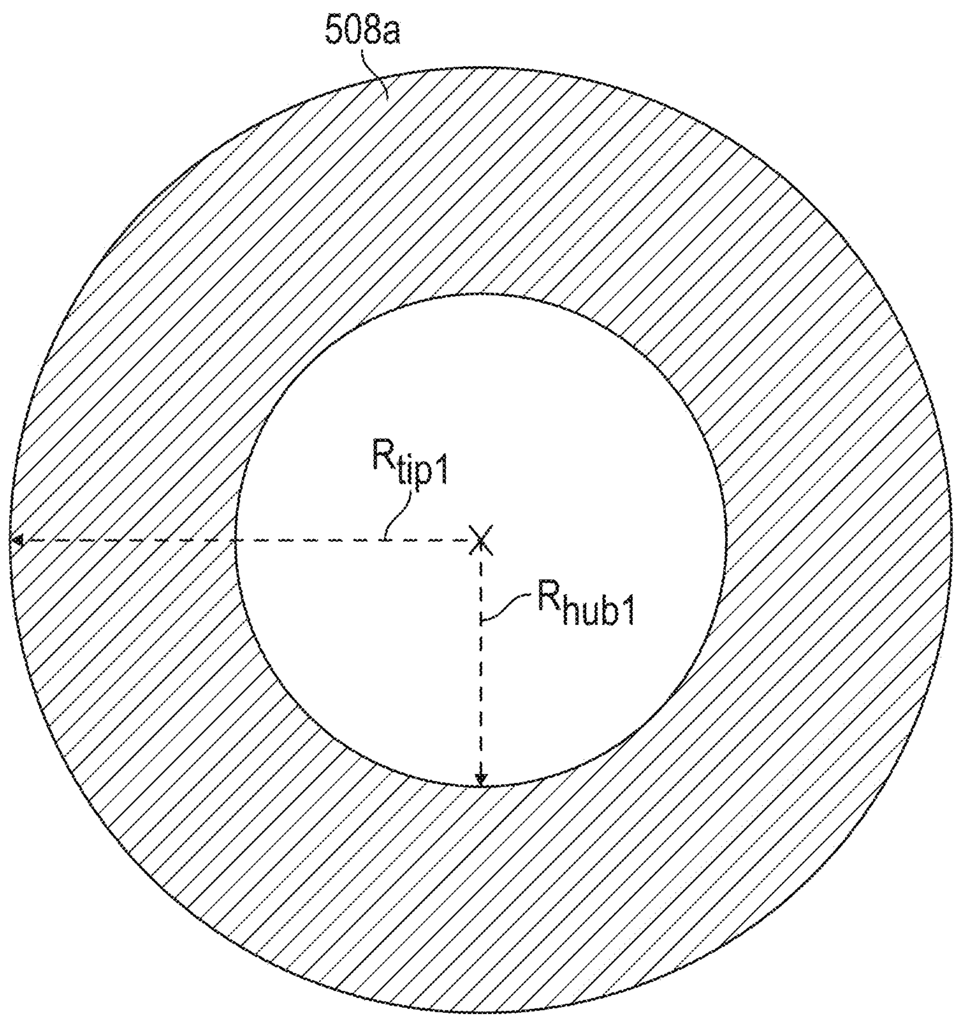
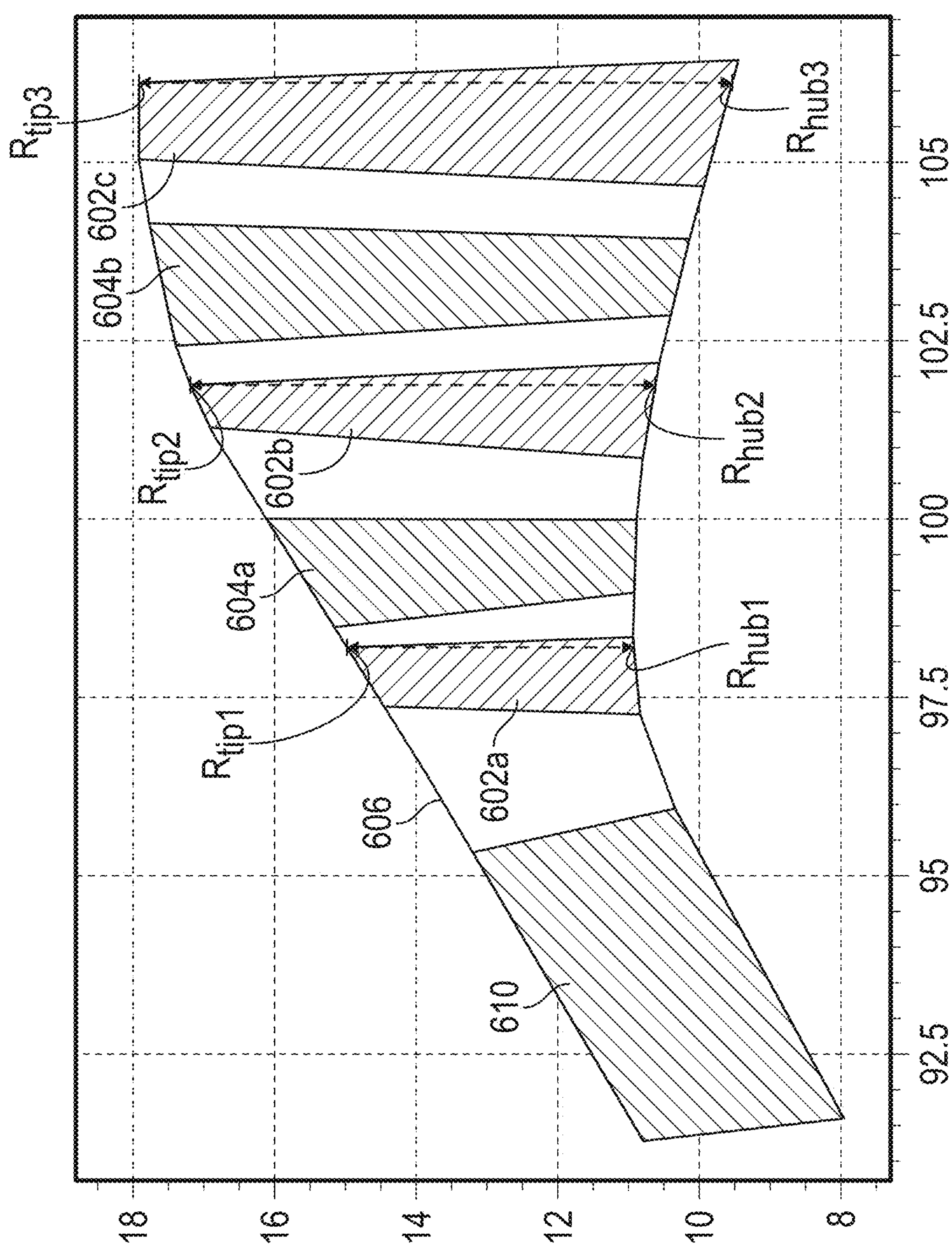


FIG. 5B



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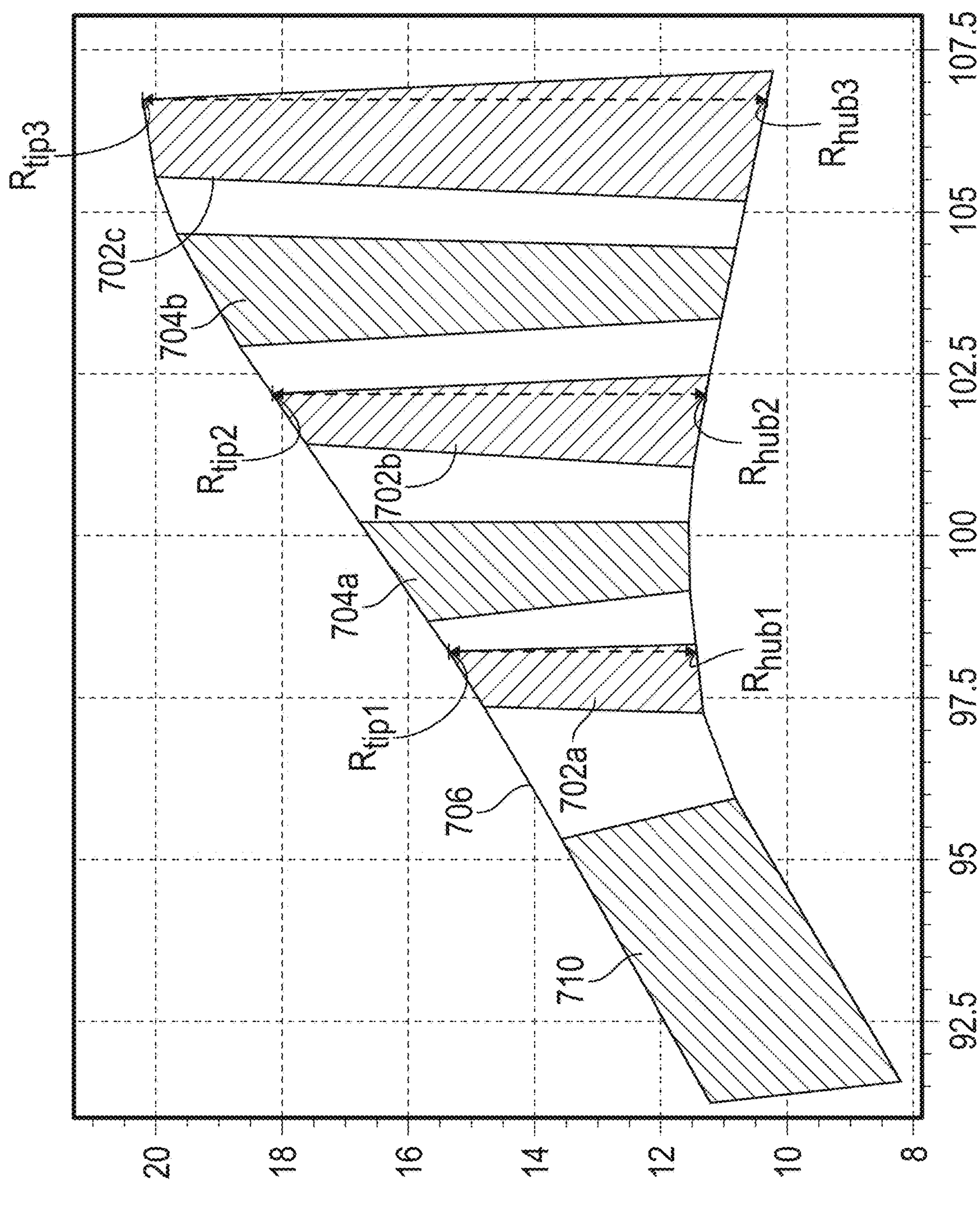
602c
R_{hub3}
R_{tip2}
602b
R_{hub2}
604a
R_{tip1}
R_{hub1}
606
602a
610
92.5
95
97.5
100
102.5
105
8
10
12
14
16
18

FIG. 6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702b
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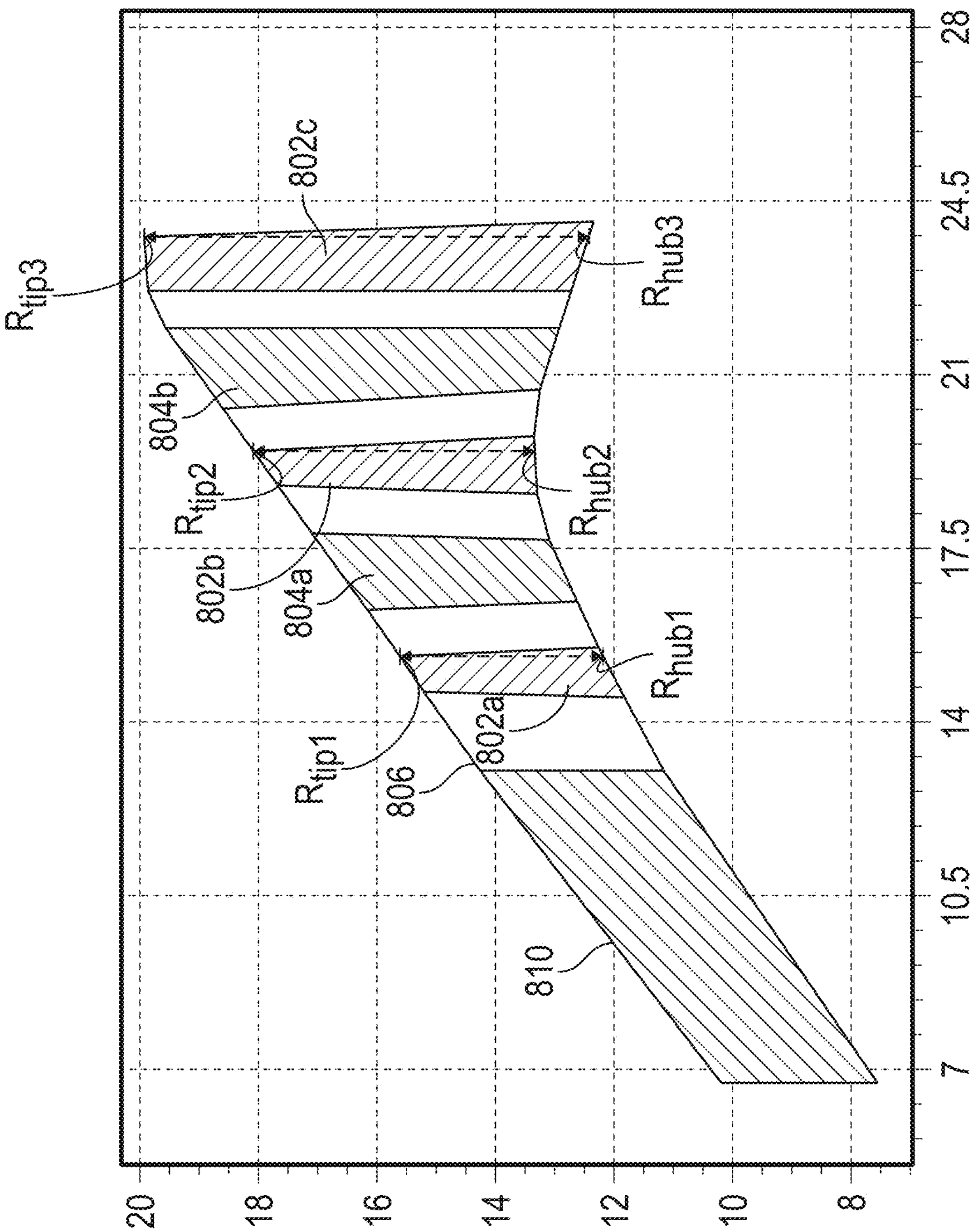
R_{tip1}
706
702a
710
R_{hub3}
R_{hub2}
R_{hub1}
20
18
16
14
12
10
8
92.5
95
97.5
100
102.5
105
107.5

FIG. 7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804b
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R_hub2
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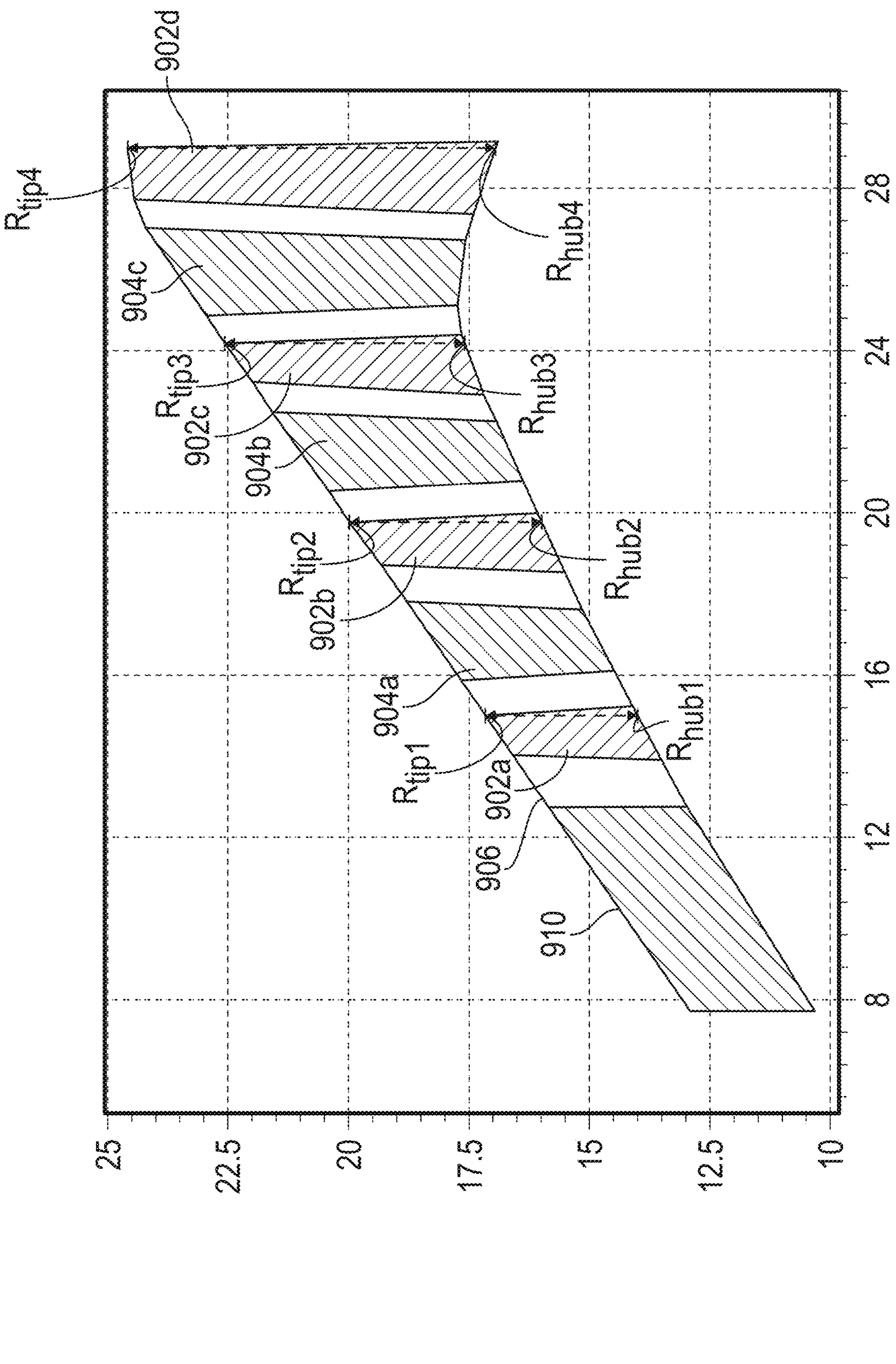
804a
R_hub1
802a
R_tip1
806
810
20
18
16
14
12
10
8
7
10.5
14
17.5
21
24.5
28

FIG. 8

	LPT STAGES (N)	GEAR RATIO	EGT @ R/L (°C)	STG1 EXIT AREA (IN^2)	STG2 EXIT AREA (IN^2)	STG3 EXIT AREA (IN^2)	AREA RATIO	AREA-EGT RATIO	STG1 AN^2 @R/L	STG3 AN^2 @R/L
ENGINE 01	3	2.95	1083	326.7	527.3	728.5	2.23	1.38	35	78
ENGINE 02	3	2.95	1083	326.7	527.3	728.5	2.23	1.38	35	78
ENGINE 03	3	3.14	1083	285.0	461.8	638.4	2.24	1.38	35	78
ENGINE 04	3	3.40	1083	326.7	525.7	725.3	2.22	1.38	36	80
ENGINE 05	3	3.40	1083	326.7	576.2	826.6	2.53	1.47	36	91
ENGINE 06	3	3.40	1083	326.7	638.2	950.7	2.91	1.58	36	105
ENGINE 07	3	3.06	1083	372.4	699.6	1027.8	2.76	1.53	33	91
ENGINE 08	3	2.84	1100	316.1	544.9	774.4	2.45	1.42	32	78
ENGINE 09	3	8.33	1067	293.2	528.8	765.3	2.61	1.51	30	78

FIG. 9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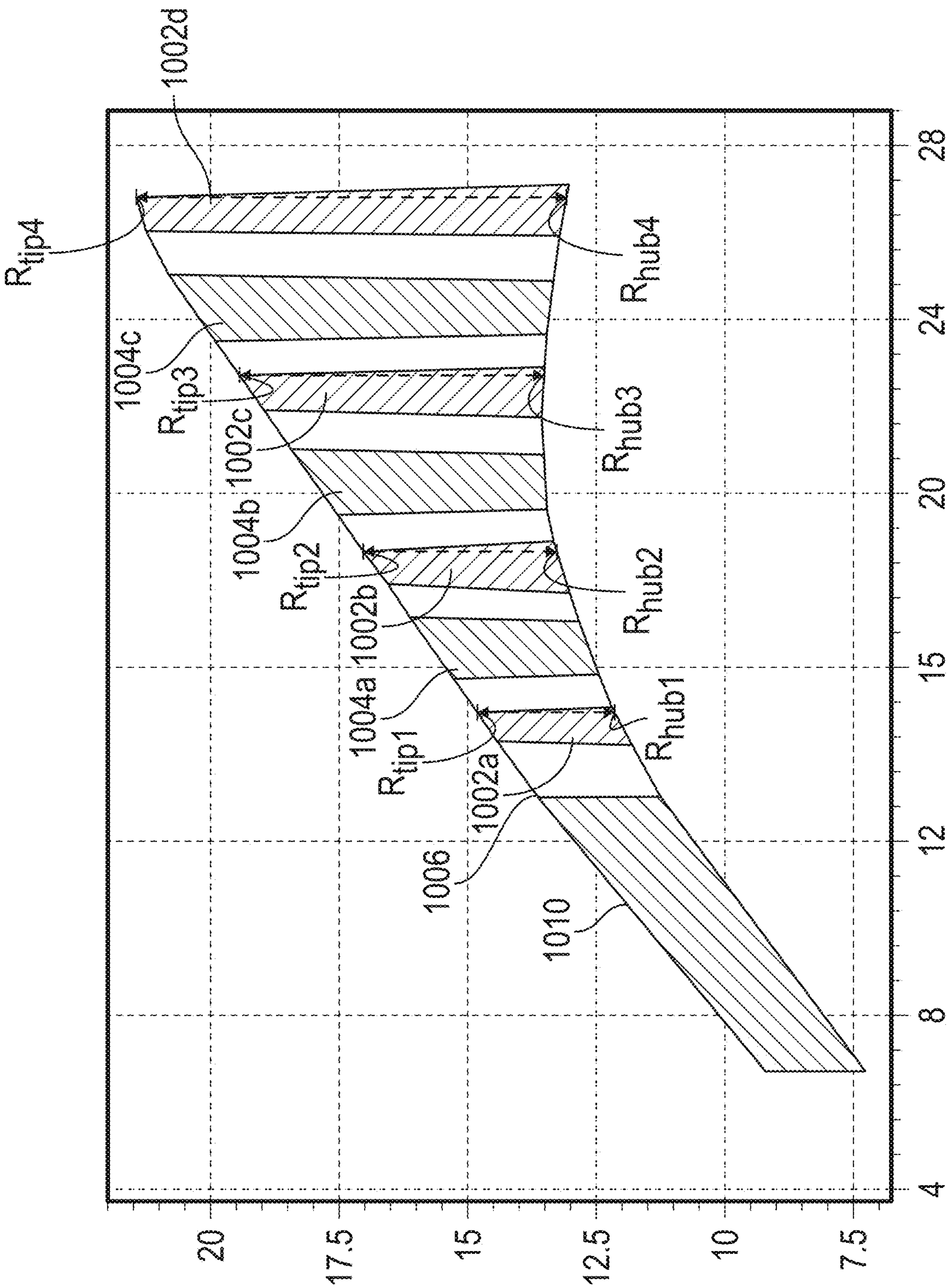
R_{hub4}
904c
R_{tip3}
R_{hub3}
902c
904b
R_{tip2}
R_{hub2}
902b
904a
R_{tip1}
R_{hub1}
902a
906
910
8
12
16
20
24
28
25
22.5
20
17.5
15
12.5
10

FIG. 10



R_{tip4}
1002d
R_{hub4}
1004c
R_{tip3}
1002c
R_{hub3}
1004b
R_{tip2}
1002b
R_{hub2}
1004a
R_{tip1}
1002a
R_{hub1}
1006
1010
4
8
12
15
20
24
28
20
17.5
15
12.5
10
7.5

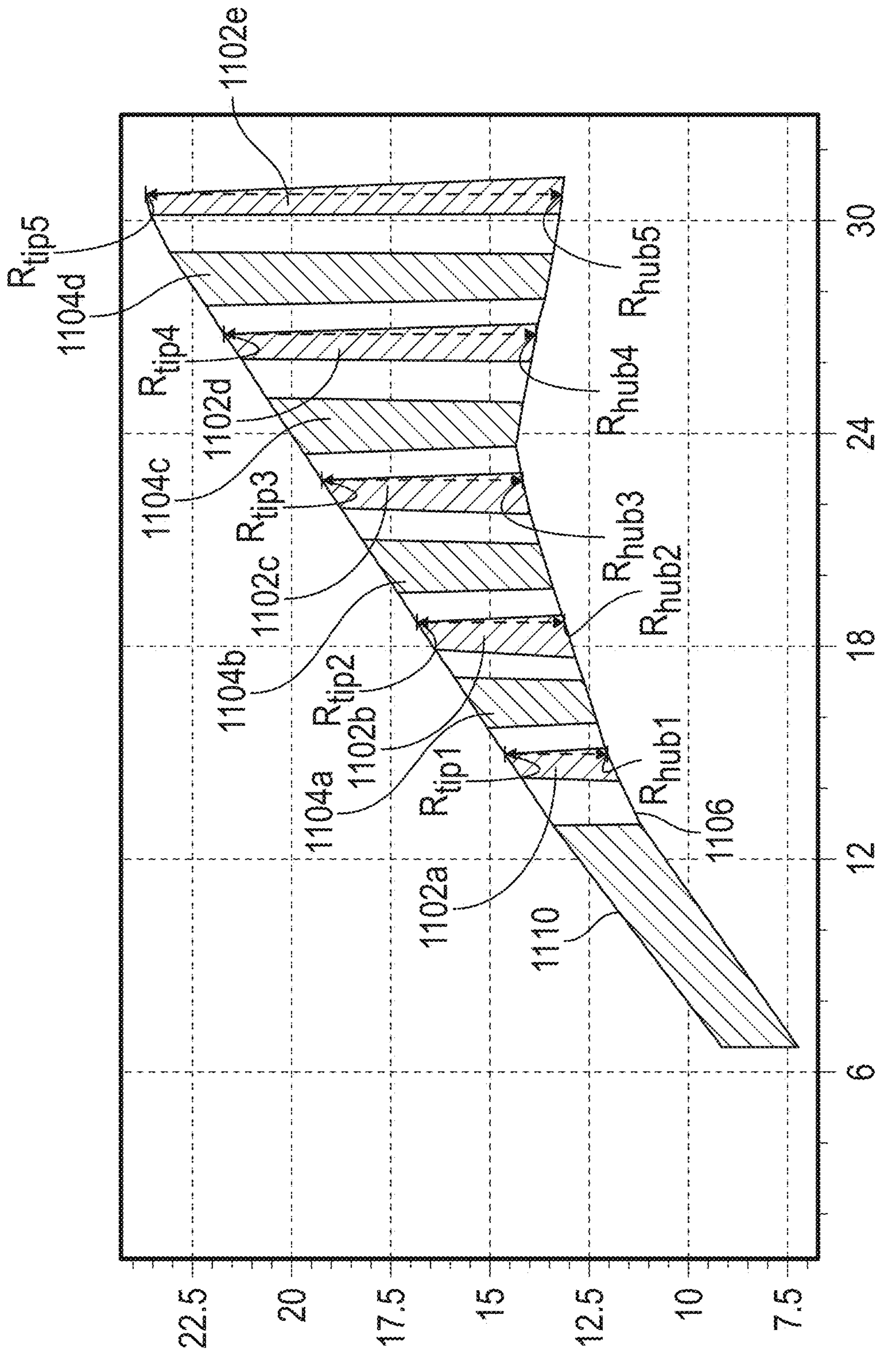
FIG. 11

	LPT STAGES (N)	GEAR RATIO	EGT @ R/L (°C)	STG1 EXIT AREA (IN^2)	STG2 EXIT AREA (IN^2)	STG3 EXIT AREA (IN^2)	STG4 EXIT AREA (IN^2)	AREA RATIO	AREA-EGT RATIO	STG1 AN^2 @R/L	STG4 AN^2 @R/L
ENGINE 10	4	2.60	1080	242.5	319.8	408.5	497.1	2.05	1.18	20	41
ENGINE 11	4	2.60	1080	261.7	345.1	440.8	536.5	2.05	1.18	21	43
ENGINE 12	4	2.60	1080	261.7	380.9	517.6	654.3	2.50	1.26	21	53
ENGINE 13	4	2.60	1080	261.7	579.5	944.1	1308.5	5.00	1.58	21	105
ENGINE 14	4	2.60	1080	261.7	452.4	671.1	889.8	3.40	1.39	21	71
ENGINE 15	4	2.50	1080	257.6	445.3	660.6	875.8	3.40	1.39	17	58
ENGINE 16	4	2.30	1080	299.2	408.2	533.2	658.2	2.20	1.20	13	29
ENGINE 17	4	8.70	1175	171.9	252.8	345.6	438.3	2.55	1.16	16	41
ENGINE 18	4	8.70	1175	171.9	297.1	440.8	584.5	3.40	1.28	16	54
ENGINE 19	4	8.70	1175	171.9	354.5	564.1	773.6	4.50	1.41	16	72
ENGINE 20	4	8.70	1175	191.0	336.0	502.3	668.5	3.50	1.29	14	49
ENGINE 21	4	7.83	1175	212.2	347.5	502.7	657.8	3.10	1.24	18	56
ENGINE 22	4	7.83	1175	222.4	367.6	534.1	700.6	3.15	1.25	20	63
ENGINE 23	4	7.83	1175	232.2	380.2	550.1	719.8	3.10	1.24	19	59
ENGINE 24	4	8.70	1175	171.9	250.2	340.0	429.8	2.50	1.16	17	43
ENGINE 25	4	7.91	1175	193.7	337.8	503.0	668.3	3.45	1.29	19	66
ENGINE 26	4	8.33	1175	175.6	308.9	461.8	614.6	3.50	1.29	15	53
ENGINE 27	4	8.33	1067	237.3	338.2	453.9	569.5	2.40	1.25	17	41
ENGINE 28	4	6.96	1067	283.0	368.9	467.5	566.0	2.00	1.18	21	42
ENGINE 29	4	8.70	1175	171.9	252.8	345.6	438.3	2.55	1.16	16	41
ENGINE 30	4	6.96	1175	244.4	326.0	419.6	513.2	2.10	1.09	16	34
ENGINE 31	4	8.33	1175	175.6	346.2	541.9	737.5	4.20	1.37	17	71
ENGINE 32	4	8.33	1067	239.7	359.8	497.5	635.2	2.65	1.30	22	58

FIG. 12

	LPT STAGES (N)	GEAR RATIO	EGT @ R/L (°C)	STG1 EXIT AREA (IN^2)	STG2 EXIT AREA (IN^2)	STG3 EXIT AREA (IN^2)	STG4 EXIT AREA (IN^2)	AREA RATIO	AREA-EGT RATIO	STG1 AN^2 @R/L	STG4 AN^2 @R/L
ENGINE 33	4	8.70	1175	171.9	213.6	261.5	309.4	1.80	1.04	25	45
ENGINE 34	4	6.96	1175	244.4	674.7	1168.4	1661.9	6.80	1.61	16	109
ENGINE 35	4	3.10	1080	261.7	674.8	1148.8	1622.5	6.20	1.70	17	105
ENGINE 36	4	2.79	1080	257.6	625.1	1046.8	1468.3	5.70	1.65	12	68
ENGINE 37	4	8.70	1175	171.9	216.3	267.1	318.0	1.85	1.04	27	50
ENGINE 38	4	6.96	1175	244.4	671.0	1160.4	1649.7	6.75	1.61	18	122
ENGINE 39	4	3.10	1080	261.7	293.5	329.9	366.4	1.40	1.04	26	36
ENGINE 40	4	2.79	1080	257.6	660.3	1122.4	1584.2	6.15	1.70	16	98

FIG. 13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1102a
1110
Rhub5
Rhub4
Rhub3
Rhub2
Rhub1
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24
30
22.5
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17.5
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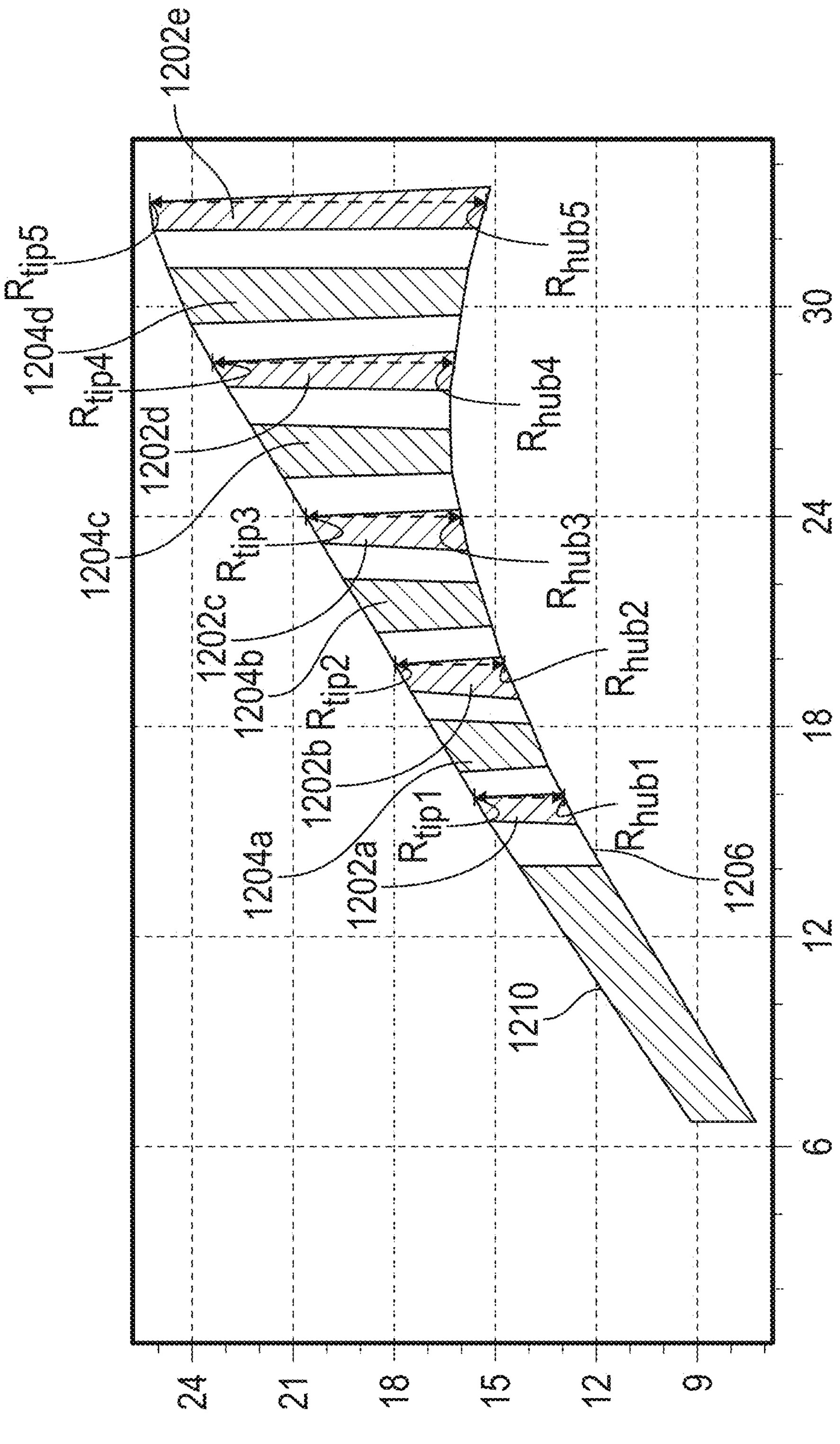
12.5
10
7.5

FIG. 14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1204a
R_{tip1}
R_{hub1}
1202a
1206
1210
6
12
18
24
30
9
12
15
18
21
24

FIG. 15

	LPT STAGES (N)	GEAR RATIO	EGT @ R/L (°C)	STG1 EXIT AREA (IN^2)	STG2 EXIT AREA (IN^2)	STG3 EXIT AREA (IN^2)	STG4 EXIT AREA (IN^2)	STG5 EXIT AREA (IN^2)	AREA RATIO	AREA-EGT RATIO	STG1 AN^2 @R/L	STG5 AN^2 @R/L
ENGINE 41	5	6.96	1175	212.1	341.6	524.5	875.0	1212.0	5.72	1.32	15	84
ENGINE 42	5	6.96	1175	220.4	331.0	508.8	874.6	1210.0	5.49	1.30	12	65
ENGINE 43	5	7.40	1175	180.5	256.9	508.7	816.6	1161.0	6.43	1.36	9	59
ENGINE 44	5	6.96	1175	232.6	326.9	527.7	895.0	1279.3	5.50	1.30	14	76
ENGINE 45	5	7.56	1175	155.4	250.2	379.3	563.1	851.4	5.48	1.30	14	76

FIG. 16

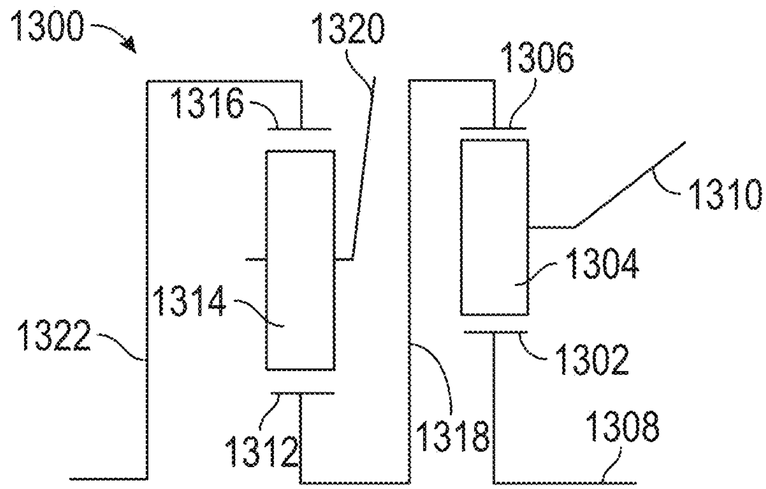
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FIG. 17

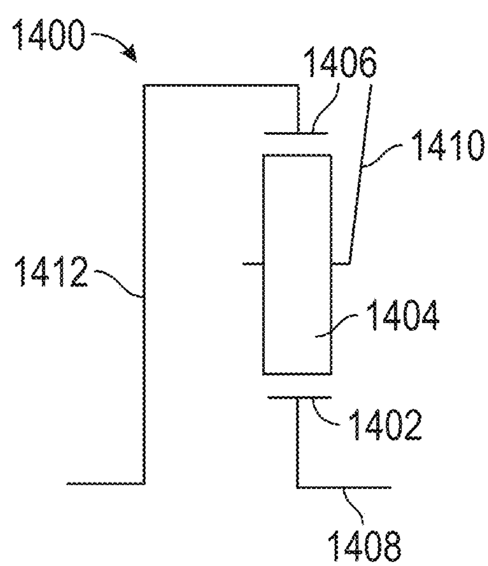
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FIG. 18

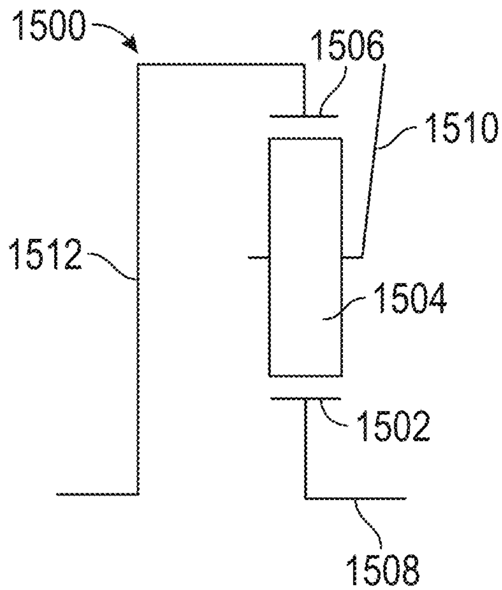
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FIG. 19

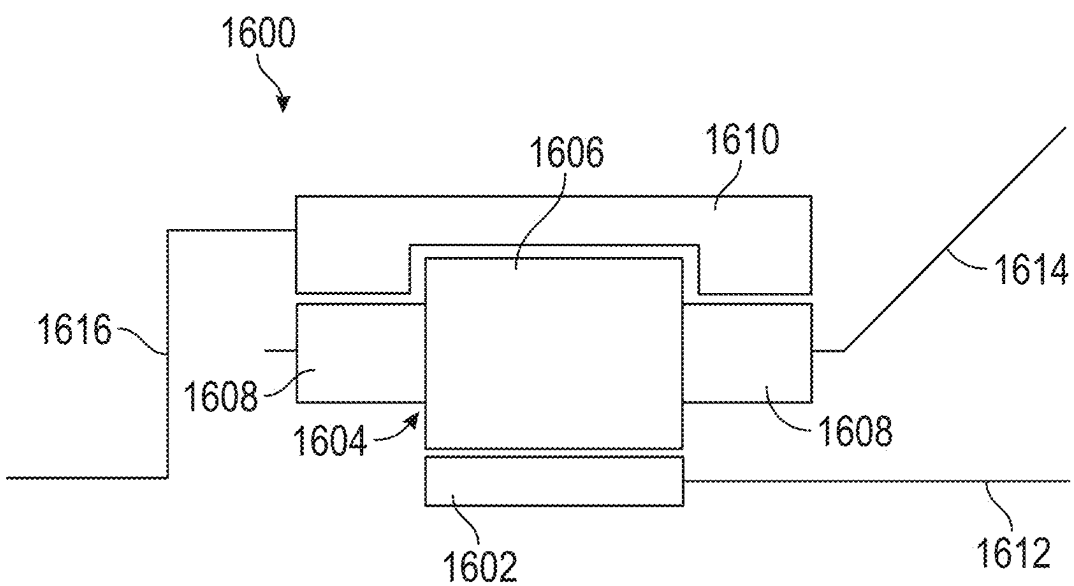
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FIG. 20

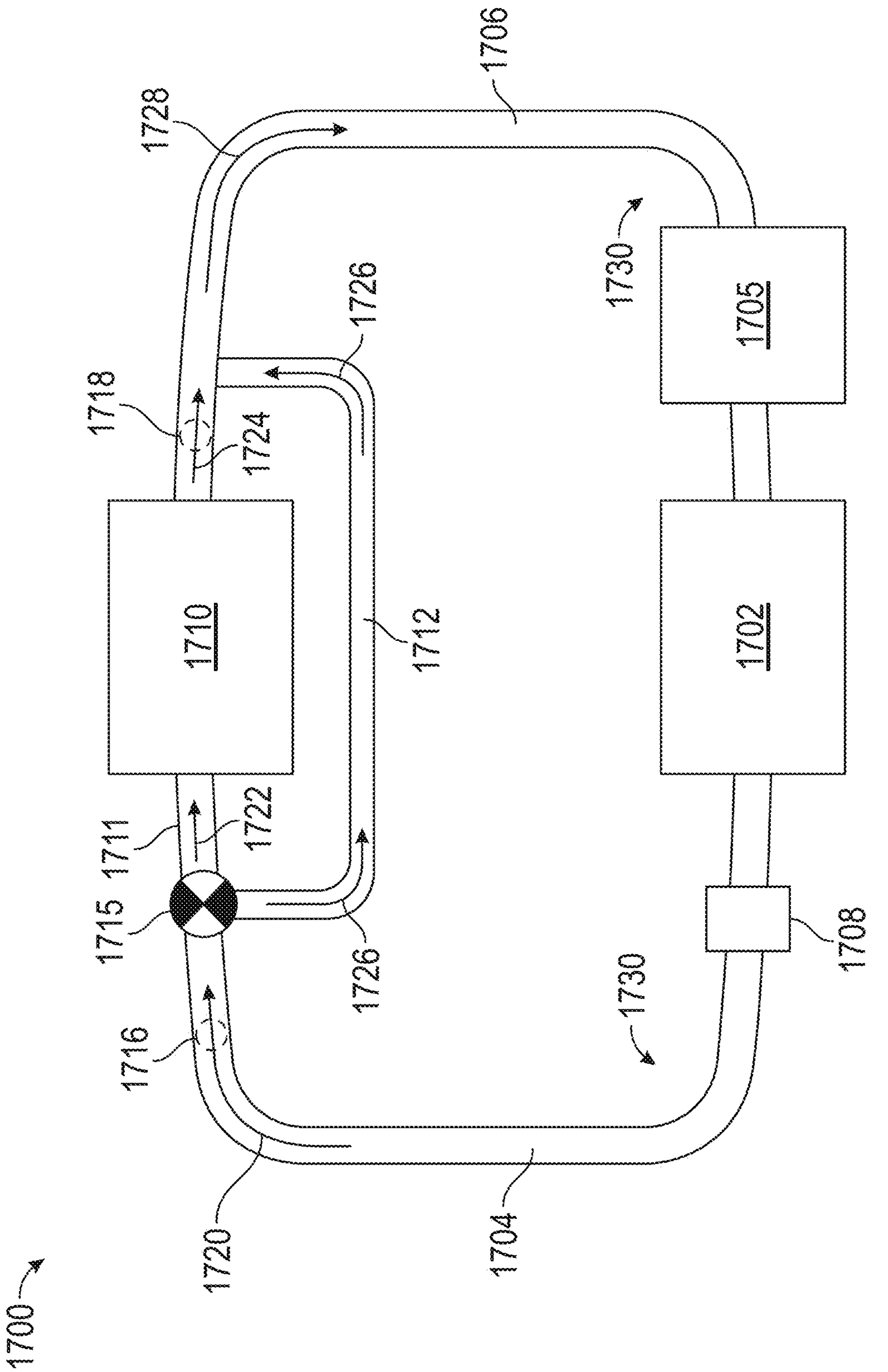
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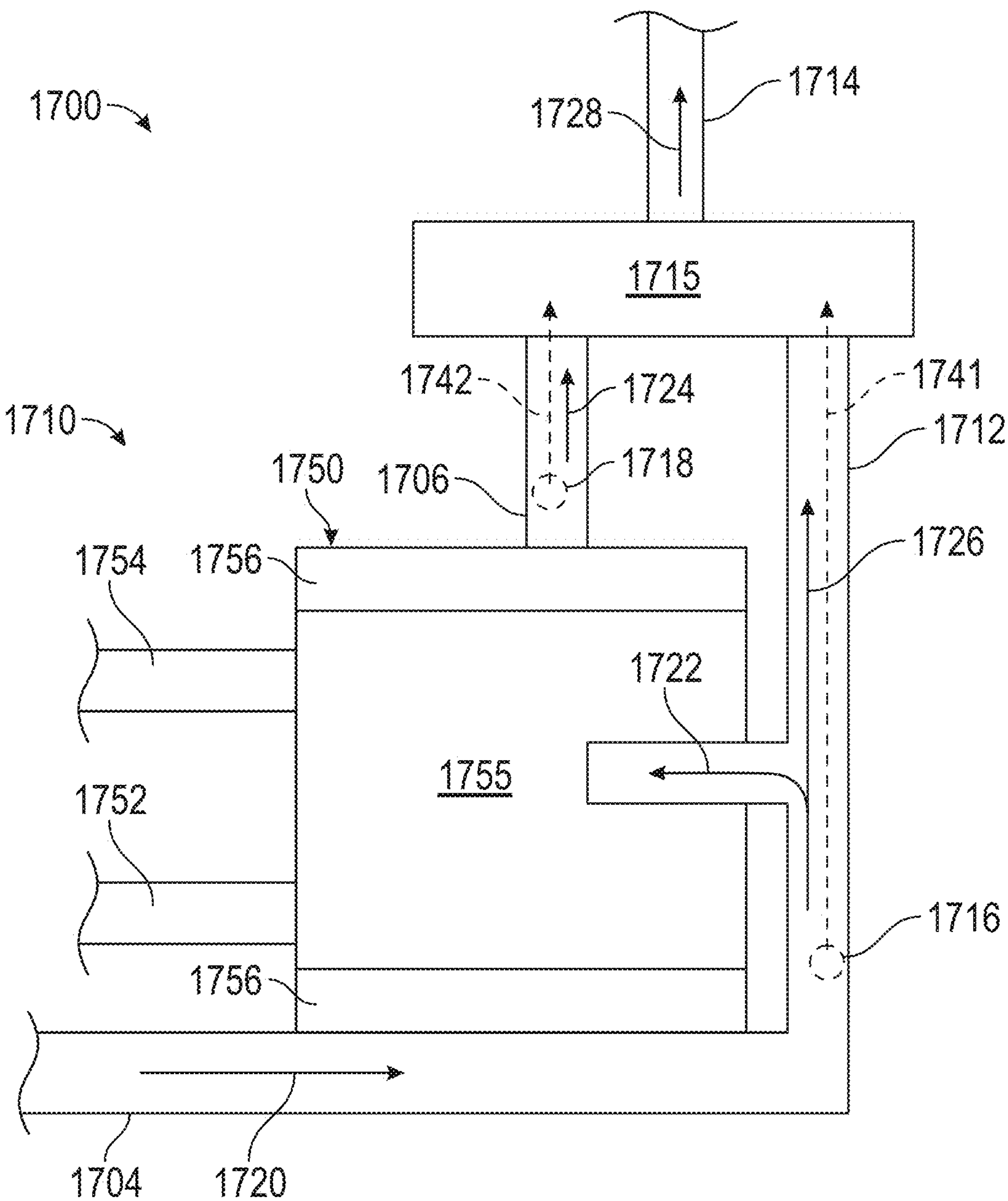
FIG. 21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1755
1752
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1756
1704
1720

FIG. 22

TURBOMACHINERY ENGINES WITH HIGH-SPEED LOW-PRESSURE TURBINES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 18/318,604, filed May 16, 2023, which claims the benefit of Indian Patent Application No. 202311010789, filed Feb. 17, 2023. The prior applications are incorporated by reference herein.

FIELD

This disclosure relates generally to turbomachinery engines comprising a gearbox and particularly to geared turbofan engines.

BACKGROUND

A turbofan engine is a type of turbomachinery engine and includes a core engine that drives a bypass fan. The bypass fan generates the majority of the thrust of the turbofan engine. The generated thrust can be used to move a payload (e.g., an aircraft).

In some instances, a turbofan engine is configured as a direct drive engine. Direct drive engines are configured such that a power turbine (e.g., a low-pressure turbine) of the core engine is directly coupled to the bypass fan. As such, the power turbine and the bypass fan rotate at the same rotational speed (i.e., the same rpm).

In other instances, a turbofan engine can be configured as a geared engine. Geared engines include a gearbox disposed between and interconnecting the bypass fan and power turbine of the core engine. The gearbox, for example, allows the power turbine of the core engine to rotate at a different speed than the bypass fan. Thus, the gearbox can, for example, allow the power turbine of the core engine and the bypass fan to operate at their respective rotational speeds for improved efficiency and/or power production.

There is an ongoing need for improved engine configurations for geared turbofan engines.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a cross-sectional schematic illustration of an example of a turbomachinery engine configured with an open rotor propulsion system, according to the present disclosure.

FIG. **2** is a cross-sectional schematic illustration of an example of a turbomachinery engine configured with an open rotor propulsion system, according to the present disclosure.

FIG. **3** is a cross-sectional schematic illustration of an example of a turbomachinery engine configured with a ducted propulsion system, according to the present disclosure.

FIG. **4** is a cross-sectional schematic illustration of an example of a turbomachinery engine configured with a ducted propulsion system, according to the present disclosure.

FIG. **5**A is a cross-sectional schematic illustration of an example of a low-pressure turbine comprising three rotating blade stages, according to the present disclosure.

FIG. **5**B is a cross-sectional schematic illustration depicting an exit area of a first rotating blade stage of the low-pressure turbine of FIG. **5**A, according to the present disclosure.

FIG. **6** is a cross-sectional schematic illustration of another example of a low-pressure turbine comprising three rotating blade stages, according to the present disclosure.

FIG. **7** is a cross-sectional schematic illustration of another example of a low-pressure turbine comprising three rotating blade stages, according to the present disclosure.

FIG. **8** is a cross-sectional schematic illustration of another example of a low-pressure turbine comprising three rotating blade stages, according to the present disclosure.

FIG. **9** is a chart depicting various engine parameters of several exemplary turbomachinery engines comprising three rotating blade stages, according to the present disclosure.

FIG. **10** is a cross-sectional schematic illustration of an example of a low-pressure turbine comprising four rotating blade stages, according to the present disclosure.

FIG. **11** is a cross-sectional schematic illustration of another example of a low-pressure turbine comprising four rotating blade stages, according to the present disclosure.

FIG. **12** is a chart depicting various engine parameters of several exemplary turbomachinery engines comprising four rotating blade stages, according to the present disclosure.

FIG. **13** is a chart depicting various engine parameters of several exemplary turbomachinery engines comprising four rotating blade stages.

FIG. **14** is a cross-sectional schematic illustration of an example of a low-pressure turbine comprising five rotating blade stages, according to the present disclosure.

FIG. **15** is a cross-sectional schematic illustration of another example of a low-pressure turbine comprising five rotating blade stages, according to the present disclosure.

FIG. **16** is a chart depicting various engine parameters of several exemplary turbomachinery engines comprising five rotating blade stages, according to the present disclosure.

FIG. **17** is a cross-sectional schematic illustration of an example of a gearbox configuration for a turbomachinery engine, according to the present disclosure.

FIG. **18** is a cross-sectional schematic illustration of an example of a gearbox configuration for a turbomachinery engine, according to the present disclosure.

FIG. **19** is a cross-sectional schematic illustration of an example of a gearbox configuration for a turbomachinery engine, according to the present disclosure.

FIG. **20** is a cross-sectional schematic illustration of an example of a gearbox configuration for a turbomachinery engine, according to the present disclosure.

FIG. **21** is a schematic diagram of an exemplary lubricant system supplying lubricant to an engine component, according to the present disclosure.

FIG. **22** is a schematic diagram of the lubricant system configured to supply lubricant to a gearbox, according to the present disclosure.

DETAILED DESCRIPTION

Reference now will be made in detail to examples of the disclosed technology, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the disclosed technology, not a limitation of the disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the scope or spirit of the disclosure. For instance, features illustrated or described as part of one example can be used with another example to yield a still further example. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.

The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any implementation described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other implementations.

As used herein, the terms "first," "second," and "third" may be used interchangeably to distinguish one component from another and are not intended to signify the location or importance of the individual components.

The terms "forward" and "aft" refer to relative positions within a gas turbine engine or vehicle and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

The terms "upstream" and "downstream" refer to the relative direction with respect to fluid flow in a fluid pathway. For example, "upstream" refers to the direction from which the fluid flows, and "downstream" refers to the direction to which the fluid flows.

The terms "coupled," "fixed," "attached to," and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

The singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise.

Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

One or more components of the turbomachinery engine or gear assembly described herein below may be manufactured or formed using any suitable process, such as an additive manufacturing process, such as a 3-D printing process. The use of such a process may allow such components to be formed integrally, as a single monolithic component, or as any suitable number of sub-components. In particular, the additive manufacturing process may allow such components to be integrally formed and include a variety of features not possible when using prior manufacturing methods. For example, the additive manufacturing methods described herein enable the manufacture of heat exchangers having unique features, configurations, thicknesses, materials, densities, fluid passageways, headers, and mounting structures that may not have been possible or practical using prior manufacturing methods. Some of these features are described herein.

Rising fuel prices, depleting natural resources, and regulatory constraints place increasing demands on turbomachinery engines. As such, turbomachinery engines with improved efficiency and performance are desired. Designing turbomachinery engines, however, is complex, time consuming, and expensive. There are many engine components and parameters to consider (each of various weight), and many are of the components and parameters are interdependent. Therefore, changing one component or one parameter can often create cascading effects requiring one or more other parameters or components to be reconfigured.

Various turbomachinery engines and gear assemblies are disclosed herein. The disclosed turbomachinery engines have improved efficiency and/or performance than typical turbomachinery engines.

The disclosed turbomachinery engines comprise a gearbox and a turbine (e.g., a low-pressure turbine) coupled to the gearbox. The disclosed turbomachinery engines are characterized or defined by one or more parameters of a turbine (e.g., the low-pressure turbine). These turbine parameters include: an area ratio and/or an area-EGT ratio. Additional information about these ratios and exemplary engines comprising these ratios are provided below.

Referring now to the drawings, FIG. **1** is an example of an engine **100** including a gear assembly **102** according to aspects of the present disclosure. The engine **100** includes a fan assembly **104** driven by a core engine **106**. In various examples, the core engine **106** is a Brayton cycle system configured to drive the fan assembly **104**. The core engine **106** is shrouded, at least in part, by an outer casing **114**. The fan assembly **104** includes a plurality of fan blades **108**. A vane assembly **110** extends from the outer casing **114** in a cantilevered manner. Thus, the vane assembly **110** can also be referred to as an unducted vane assembly. The vane assembly **110**, including a plurality of vanes **112**, is positioned in operable arrangement with the fan blades **108** to provide thrust, control thrust vector, abate or re-direct undesired acoustic noise, and/or otherwise desirably alter a flow of air relative to the fan blades **108**.

In some examples, the fan assembly **104** includes eight (8) to twenty-two (22) fan blades **108**. In particular examples, the fan assembly **104** includes ten (10) to eighteen (18) fan blades **108**. In certain examples, the fan assembly **104** includes twelve (12) to sixteen (16) fan blades **108**. In some examples, the vane assembly **110** includes three (3) to thirty (30) vanes **112**. In certain examples, the vane assembly **110** includes an equal or fewer quantity of vanes **112** to fan blades **108**. For example, in particular examples, the engine **100** includes twelve (12) fan blades **108** and ten (10) vanes **112**. In other examples, the vane assembly **110** includes a greater quantity of vanes **112** to fan blades **108**. For example, in particular implementations, the engine **100** includes ten (10) fan blades **108** and twenty-three (23) vanes **112**.

In certain examples, such as depicted in FIG. **1**, the vane assembly **110** is positioned downstream or aft of the fan assembly **104**. However, it should be appreciated that in some examples, the vane assembly **110** may be positioned upstream or forward of the fan assembly **104**. In still various examples, the engine **100** may include a first vane assembly positioned forward of the fan assembly **104** and a second vane assembly positioned aft of the fan assembly **104**. The fan assembly **104** may be configured to desirably adjust pitch at one or more fan blades **108**, such as to control thrust vector, abate or re-direct noise, and/or alter thrust output. The vane assembly **110** may be configured to desirably adjust pitch at one or more vanes **112**, such as to control thrust vector, abate or re-direct noise, and/or alter thrust output. Pitch control mechanisms at one or both of the fan assembly **104** or the vane assembly **110** may co-operate to produce one or more desired effects described above.

In certain examples, such as depicted in FIG. **1**, the engine **100** is an un-ducted thrust producing system, such that the plurality of fan blades **108** is unshrouded by a nacelle or fan casing. As such, in various examples, the engine **100** may be configured as an unshrouded turbofan engine, an open rotor engine, or a propfan engine. In particular examples, the engine **100** is an unducted rotor engine with a single row of fan blades **108**. The fan blades **108** can have a large diameter, such as may be suitable for high bypass ratios, high cruise speeds (e.g., comparable to aircraft with turbofan engines, or generally higher cruise speed than aircraft with turboprop engines), high cruise altitude (e.g., comparable to aircraft with turbofan engines, or generally higher cruise speed than aircraft with turboprop engines), and/or relatively low rotational speeds.

The fan blades **108** comprise a diameter (D_{fan}). It should be noted that for purposes of illustration only half of the D_{fan} is shown (i.e., the radius of the fan). In some examples, the D_{fan} is 72-216 inches. In particular examples the D_{fan} is 100-200 inches. In certain examples, the D_{fan} is 120-190 inches. In other examples, the D_{fan} is 72-120 inches. In some examples, the D_{fan} is 80-90 inches. In yet other examples, the D_{fan} is 50-80 inches.

In some examples, the fan blade tip speed at a cruise flight condition can be 650 to 1000 fps, or 800 to 900 fps. A fan pressure ratio (FPR) for the fan assembly **104** can be 1.04 to 1.10, or in some examples 1.05 to 1.08, as measured across the fan blades at a cruise flight condition. In other examples, the FPR can be within a range of 1.04-1.8, 1.1-1.4, 1.3-1.6, or 1.5-1.8.

Cruise altitude is generally an altitude at which an aircraft levels after climb and prior to descending to an approach flight phase. In various examples, the engine is applied to a vehicle with a cruise altitude up to approximately 65,000 ft. In certain examples, cruise altitude is from approximately 28,000 ft. to approximately 45,000 ft. In still certain examples, cruise altitude is expressed in flight levels (FL) based on standard air pressure at sea level, in which a cruise flight condition is from FL280 to FL650. In another example, cruise flight condition is from FL280 to FL450. In still certain examples, cruise altitude is defined based at least on barometric pressure, in which cruise altitude is from approximately 4.85 psia to approximately 0.82 psia based on a sea-level pressure of approximately 14.70 psia and sea-level temperature at approximately 59 degrees Fahrenheit. In another example, cruise altitude is from approximately 4.85 psia to approximately 2.14 psia. It should be appreciated that in certain examples, the ranges of cruise altitude defined by pressure may be adjusted based on a different reference sea-level pressure and/or sea-level temperature.

The core engine **106** is generally encased in outer casing **114** defining one-half of a core diameter (D_{core}), which may be thought of as the maximum extent from the centerline axis (datum for R). In certain examples, the engine **100** includes a length (L) from a longitudinally (or axial) forward end **116** to a longitudinally aft end **118**. In various examples, the engine **100** defines a ratio of L/D_{core} that provides for reduced installed drag. In one example, L/D_{core} is at least 2. In another example, L/D_{core} is at least 2.5. In some examples, the L/D_{core} is less than 5, less than 4, and less than 3. In various examples, it should be appreciated that the L/D_{core} is for a single unducted rotor engine.

The reduced installed drag may further provide for improved efficiency, such as improved specific fuel consumption. Additionally, or alternatively, the reduced installed drag may provide for cruise altitude engine and aircraft operation at or above Mach 0.5. In certain examples, the L/D_{core}, the fan assembly **104**, and/or the vane assembly **110** separately or together configure, at least in part, the engine **100** to operate at a maximum cruise altitude operating speed from approximately Mach 0.55 to approximately Mach 0.85; or from approximately Mach 0.72 to Mach 0.85 or from approximately Mach 0.75 to Mach 0.85.

Referring still to FIG. **1**, the core engine **106** extends in a radial direction (R) relative to an engine centerline axis **120**. The gear assembly **102** receives power or torque from the core engine **106** through a power input source **122** and provides power or torque to drive the fan assembly **104**, in a circumferential direction C about the engine centerline axis **120**, through a power output source **124**.

The gear assembly **102** of the engine **100** can include a plurality of gears, including an input and an output. The gear assembly **102** can also include one or more intermediate gears disposed between and/or interconnecting the input and the output. The input can be coupled to a turbine section of the core engine **106** and can comprise a first rotational speed. The output can be coupled to the fan assembly **104** and can have a second rotational speed. In some examples, a gear ratio of the first rotational speed to the second rotational speed is less than or equal to four (e.g., within a range of 2.0-4.0). In other examples, a gear ratio of the first rotational speed to the second rotational speed is greater than four (e.g., within a range of 4.1-14.0).

The gear assembly **102** (which can also be referred to as "a gearbox") can comprise various types and/or configurations. For example, in some instances, the gearbox is an epicyclic gearbox configured in a star gear configuration. Star gear configurations comprise a sun gear, a plurality of star gears (which can also be referred to as "planet gears"), and a ring gear. The sun gear is the input and is coupled to the power turbine (e.g., the low-pressure turbine) such that the sun gear and the power turbine rotate at the same rotational speed. The star gears are disposed between and interconnect the sun gear and the ring gear. The star gears are rotatably coupled to a fixed carrier. As such, the star gears can rotate about their respective axes but cannot collectively orbit relative to the sun gear or the ring gear. As another example, the gearbox is an epicyclic gearbox configured in a planet gear configuration. Planet gear configurations comprise a sun gear, a plurality of planet gears, and a ring gear. The sun gear is the input and is coupled to the power turbine. The planet gears are disposed between and interconnect the sun gear and the ring gear. The planet gears are rotatably coupled to a rotatable carrier. As such, the planet gears can rotate about their respective axes and also collectively rotate together with the carrier relative to the sun gear and the ring gear. The carrier is the output and is coupled to the fan assembly. The ring gear is fixed from rotation.

In some examples, the gearbox is a single-stage gearbox (e.g., FIGS. **18**-**19**). In other examples, the gearbox is a multi-stage gearbox (e.g., FIGS. **17** and **20**). In some examples, the gearbox is an epicyclic gearbox. In some examples, the gearbox is a non-epicyclic gearbox (e.g., a compound gearbox-FIG. **20**).

As noted above, the gear assembly can be used to reduce the rotational speed of the output relative to the input. In some examples, a gear ratio of the input rotational speed to the output rotational speed is within a range of 2-4. For example, the gear ratio can be 2-2.9, 3.2-4, or 3.25-3.75). In some examples, a gear ratio of the input rotational speed to the output rotational speed is greater than 4.1. For example, in particular instances, the gear ratio is within a range of 4.1-14.0, within a range of 4.5-14.0, or within a range of 6.0-14.0. In certain examples, the gear ratio is within a range of 4.5-12 or within a range of 6.0-11.0. As such, in some examples, the fan assembly can be configured to rotate at a rotational speed of 800-1500 rpm at a cruise flight condition, while the power turbine (e.g., the low-pressure turbine) is configured to rotate at a rotational speed of 2,500-15,000 rpm at a cruise flight condition. In particular examples, the fan assembly can be configured to rotate at a rotational speed of 850-1350 rpm at a cruise flight condition, while the power turbine is configured to rotate at a rotational speed of 5,000-10,000 rpm at a cruise flight condition.

Various gear assembly configurations are depicted schematically in FIGS. **17-20**. These gearboxes can be used with any of the engines disclosed herein, including the engine **100**. Additional details regarding the gearboxes are provided below.

FIG. **2** shows a cross-sectional view of an engine **200**, which is configured as an example of an open rotor propulsion engine. The engine **200** is generally similar to the engine **100**, therefore, like parts will be identified with like numerals increased to the **200** series, with it being understood that the description of the like parts of the engine **100** applies to the engine **200** unless otherwise noted. For example, the gear assembly of the engine **100** is numbered "**102**" and the gear assembly of the engine **200** is numbered "**202**," and so forth. In addition to the gear assembly **202**, the engine **200** comprises a fan assembly **204** that includes a plurality of fan blades **208** distributed around the engine centerline axis **220**. Fan blades **208** are circumferentially arranged in an equally spaced relation around the engine centerline axis **220**, and each fan blade **208** has a root **225** and a tip **226**, and an axial span defined therebetween, as well as a central blade axis **228**.

The core engine **206** includes a compressor section **230**, a combustion section **232**, and a turbine section **234** (which may be referred to as "an expansion section") together in a serial flow arrangement. The core engine **206** extends circumferentially relative to an engine centerline axis **220**. The core engine **206** includes a high-speed spool that includes a high-pressure compressor **236** and a high-speed turbine **238** operably rotatably coupled together by a high-speed shaft **240**. The combustion section **232** is positioned between the high-pressure compressor **236** and the high-pressure turbine **238**.

The combustion section **232** may be configured as a deflagrative combustion section, a rotating detonation combustion section, a pulse detonation combustion section, and/or other appropriate heat addition system. The combustion section **232** may be configured as one or more of a rich-burn system or a lean-burn system, or combinations thereof. In still various examples, the combustion section **232** includes an annular combustor, a can combustor, a cannular combustor, a trapped vortex combustor (TVC), or another appropriate combustion system, or combinations thereof.

The core engine **206** also includes a booster or low-pressure compressor **242** positioned in flow relationship with the high-pressure compressor **236**. The low-pressure compressor **242** is rotatably coupled with the low-pressure turbine **244** via a low-speed shaft **246** to enable the low-pressure turbine **244** to drive the low-pressure compressor **242**. The low-speed shaft **246** is also operably connected to the gear assembly **202** to provide power to the fan assembly **204**, such as described further herein.

It should be appreciated that the terms "low" and "high," or their respective comparative degrees (e.g., "lower" and "higher", where applicable), when used with compressor, turbine, shaft, or spool components, each refer to relative pressures and/or relative speeds within an engine unless otherwise specified. For example, a "low spool" or "low-speed shaft" defines a component configured to operate at a rotational speed, such as a maximum allowable rotational speed, lower than a "high spool" or "high-speed shaft" of the engine. Alternatively, unless otherwise specified, the aforementioned terms may be understood in their superlative degree. For example, a "low turbine" or "low-speed turbine" may refer to the lowest maximum rotational speed turbine within a turbine section, a "low compressor" or "low speed compressor" may refer to the lowest maximum rotational speed compressor within a compressor section, a "high turbine" or "high-speed turbine" may refer to the highest maximum rotational speed turbine within the turbine section, and a "high compressor" or "high-speed compressor" may refer to the highest maximum rotational speed compressor within the compressor section. Similarly, the low-speed spool refers to a lower maximum rotational speed than the high-speed spool. It should further be appreciated that the terms "low" or "high" in such aforementioned regards may additionally, or alternatively, be understood as relative to minimum allowable speeds, or minimum or maximum allowable speeds relative to normal, desired, steady state, etc. operation of the engine.

The compressors and/or turbines disclosed herein can include various stage counts. As disclosed herein the stage count includes the number of rotors or blade stages in a particular component (e.g., a compressor or turbine). For example, in some instances, a low-pressure compressor (which can also be referred to as "a booster") can comprise 1-8 stages, a high-pressure compressor can comprise 8-15 stages, a high-pressure turbine comprises 1-2 stages, and/or a low-pressure turbine comprises 3-4 stages (including exactly 3 or 4 stages). For example, in certain examples, an engine can comprise a one stage low-pressure compressor, an 11 stage high-pressure compressor, a two stage high-pressure turbine, and a 7 stage low-pressure turbine. As another example, an engine can comprise a three stage low-pressure compressor, a 10 stage high-pressure compressor, a two stage high-pressure turbine, and a 7 stage low-pressure turbine. As another example, an engine can comprise a three stage low-pressure compressor, a 10 stage high-pressure compressor, a two stage high-pressure turbine, and a three stage low-pressure turbine. As another example, an engine can comprise a four stage low-pressure compressor, a 10 stage high-pressure compressor, a one stage high-pressure turbine, and a three stage low-pressure turbine. As another example, an engine can comprise a three stage low-pressure compressor, a 10 stage high-pressure compressor, a two stage high-pressure turbine, and a four stage low-pressure turbine. As another example, an engine can comprise a four stage low-pressure compressor, a 10 stage high-pressure compressor, a one stage high-pressure turbine, and a four stage low-pressure turbine. In other examples, an engine can comprise a 1-3 stage low-pressure compressor, an 8-11 stage high-pressure compressor, a 1-2 stage high-pressure turbine, and a 3-4 stage low-pressure turbine. In some examples, an engine can be configured without a low-pressure compressor.

In some examples, a low-pressure turbine is a counter-rotating low-pressure turbine comprising inner blade stages and outer blade stages. The inner blade stages extend radially outwardly from an inner shaft, and the outer blade stages extend radially inwardly from an outer drum. In particular examples, the counter-rotating low-pressure turbine comprises three inner blade stages and three outer blade stages, which can collectively be referred to as a six stage low-pressure turbine. In other examples, the counter-rotating low-pressure turbine comprises four inner blade stages and three outer blade stages, which can collectively be referred to as a seven stage low-pressure turbine.

As discussed in more detail below, the core engine **206** includes the gear assembly **202** that is configured to transfer power from the turbine section **234** and reduce an output rotational speed at the fan assembly **204** relative to the low-pressure turbine **244**. Examples of the gear assembly **202** depicted and described herein can allow for gear ratios suitable for large diameter unducted fans (e.g., gear ratios of 4.1-14.0, 4.5-14.0, and/or 6.0-14.0). Additionally, examples of the gear assembly **202** provided herein may be suitable within the radial or diametrical constraints of the core engine **206** within an engine core cowl **272**.

Various gearbox configurations are depicted schematically in FIGS. **17-20**. These gearboxes can be used in any of the engines disclosed herein, including the engine **200**. Additional details regarding the gearboxes are provided below.

Engine **200** also includes a vane assembly **210** comprising a plurality of vanes **212** disposed around engine centerline axis **220**. Each vane **212** has a root **248** and a tip **250**, and a span defined therebetween. Vanes **212** can be arranged in a variety of manners. In some examples, they are not all equidistant from the rotating assembly.

In some examples, vanes **212** are mounted to a stationary frame and do not rotate relative to the engine centerline axis **220** but may include a mechanism for adjusting their orientation relative to their axis **254** and/or relative to the fan blades **208**. For reference purposes, FIG. **2** depicts a forward direction denoted with arrow F, which in turn defines the forward and aft portions of the system.

As depicted in FIG. **2**, the fan assembly **204** is located forward of the core engine **106** with the exhaust **256** located aft of core engine **206** in a "puller" configuration. Other configurations are possible and contemplated as within the scope of the present disclosure, such as what may be termed a "pusher" configuration where the engine core is located forward of the fan assembly. The selection of "puller" or "pusher" configurations may be made in concert with the selection of mounting orientations with respect to the airframe of the intended aircraft application, and some may be structurally or operationally advantageous depending upon whether the mounting location and orientation are wing-mounted, fuselage-mounted, or tail-mounted configurations.

Left- or right-handed engine configurations, useful for certain installations in reducing the impact of multi-engine torque upon an aircraft, can be achieved by mirroring the airfoils (e.g., **208**, **212**) such that the fan assembly **204** rotates clockwise for one propulsion system and counter-clockwise for the other propulsion system. Alternatively, an optional reversing gearbox can be provided to permit a common gas turbine core and low-pressure turbine to be used to rotate the fan blades either clockwise or counter-clockwise, i.e., to provide either left- or right-handed configurations, as desired, such as to provide a pair of oppositely-rotating engine assemblies can be provided for certain aircraft installations while eliminating the need to have internal engine parts designed for opposite rotation directions.

The engine **200** also includes the gear assembly **202** which includes a gear set for decreasing the rotational speed of the fan assembly **204** relative to the low-pressure turbine **244**. In operation, the rotating fan blades **208** are driven by the low-pressure turbine **244** via gear assembly **202** such that the fan blades **208** rotate around the engine centerline axis **220** and generate thrust to propel the engine **200**, and hence an aircraft on which it is mounted, in the forward direction F.

In some examples, a gear ratio of the input rotational speed to the output rotational speed is greater than or equal to 4.1. In particular examples, the gear ratio is within a range of 4.1-14.0, within a range of 4.5-14.0, or within a range of 6.0-14.0. In certain examples, the gear ratio is within a range of 4.5-12 or within a range of 6.0-11.0. As such, in some examples, the fan assembly can be configured to rotate at a rotational speed of 800-1500 rpm at a cruise flight condition, while the power turbine (e.g., the low-pressure turbine) is configured to rotate at a rotational speed of 5,000-10,000 rpm at a cruise flight condition. In particular examples, the fan assembly can be configured to rotate at a rotational speed of 850-1350 rpm at a cruise flight condition, while the power turbine is configured to rotate at a rotational speed of 5,500-9,500 rpm a cruise flight condition.

It may be desirable that either or both of the fan blades **208** or the vanes **212** incorporate a pitch change mechanism such that the blades can be rotated with respect to an axis of pitch rotation (annotated as **228** and **254**, respectively) either independently or in conjunction with one another. Such pitch change can be utilized to vary thrust and/or swirl effects under various operating conditions, including to provide a thrust reversing feature which may be useful in certain operating conditions such as upon landing an aircraft.

Vanes **212** can be sized, shaped, and configured to impart a counteracting swirl to the fluid so that in a downstream direction aft of both fan blades **208** and vanes **212** the fluid has a greatly reduced degree of swirl, which translates to an increased level of induced efficiency. Vanes **212** may have a shorter span than fan blades **208**, as shown in FIG. **2**. For example, vanes **212** may have a span that is at least 50% of a span of fan blades **208**. In some examples, the span of the vanes can be the same or longer than the span as fan blades **208**, if desired. Vanes **212** may be attached to an aircraft structure associated with the engine **200**, as shown in FIG. **2**, or another aircraft structure such as a wing, pylon, or fuselage. Vanes **212** may be fewer or greater in number than, or the same in number as, the number of fan blades **208**. In some examples, the number of vanes **212** are greater than two, or greater than four, in number. Fan blades **208** may be sized, shaped, and contoured with the desired blade loading in mind.

In the example shown in FIG. **2**, an annular 360-degree inlet **258** is located between the fan assembly **204** and the vane assembly **210** and provides a path for incoming atmospheric air to enter the core engine **206** radially inwardly of at least a portion of the vane assembly **210**. Such a location may be advantageous for a variety of reasons, including management of icing performance as well as protecting the inlet **258** from various objects and materials as may be encountered in operation.

In the example of FIG. **2**, in addition to the open rotor or unducted fan assembly **204** with its plurality of fan blades **208**, an optional ducted fan assembly **260** is included behind fan assembly **204**, such that the engine **200** includes both a ducted and an unducted fan which both serve to generate thrust through the movement of air at atmospheric temperature without passage through the core engine **206**. The ducted fan assembly **260** is shown at about the same axial location as the vane **212**, and radially inward of the root **248** of the vane **212**. Alternatively, the ducted fan assembly **260** may be between the vane **212** and a core duct **262** or be farther forward of the vane **212**. The ducted fan assembly **260** may be driven by the low-pressure turbine **244**, or by any other suitable source of rotation, and may serve as the first stage of the low-pressure compressor **242** or may be operated separately. Air entering the inlet **258** flows through an inlet duct **264** and then is divided such that a portion flows through the core duct **262** and another portion flows through a fan duct **266**. Fan duct **266** may incorporate heat exchangers **268** and exhausts to the atmosphere through an independent fixed or variable nozzle **270** aft of the vane assembly **210** at the aft end of the fan cowl **252** and outside of the engine core cowl **272**. Air flowing through the fan duct **266** thus "bypasses" the core of the engine and does not pass through the core engine **106**.

Thus, in the example, engine **200** includes an unducted fan formed by the fan blades **208**, followed by the ducted fan assembly **260**, which directs airflow into two concentric or non-concentric ducts **262** and **266**, thereby forming a three-stream engine architecture with three paths for air which passes through the fan assembly **204**.

In the example shown in FIG. **2**, a slidable, moveable, and/or translatable plug nozzle **274** with an actuator may be included in order to vary the exit area of the nozzle **270**. A plug nozzle is typically an annular, symmetrical device that regulates the open area of an exit such as a fan stream or core stream by axial movement of the nozzle such that the gap between the nozzle surface and a stationary structure, such as adjacent walls of a duct, varies in a scheduled fashion thereby reducing or increasing a space for airflow through the duct. Other suitable nozzle designs may be employed as well, including those incorporating thrust reversing functionality. Such an adjustable, moveable nozzle may be designed to operate in concert with other systems such as variable bleed valves (VBVs), variable stator vanes (VSVs), or blade pitch mechanisms and may be designed with failure modes such as fully-open, fully-closed, or intermediate positions so that the nozzle **270** has a consistent "home" position to which it returns in the event of any system failure, which may prevent commands from reaching the nozzle **270** and/or its actuator.

In some examples, a mixing device **276** can be included in a region aft of a core nozzle **278** to aid in mixing the fan stream and the core stream to improve acoustic performance by directing core stream outward and fan stream inward.

Since the engine **200** shown in FIG. **2** includes both an open rotor fan assembly **204** and a ducted fan assembly **260**, the thrust output of both and the work split between them can be tailored to achieve specific thrust, fuel burn, thermal management, and/or acoustic signature objectives which may be superior to those of a typical ducted fan gas turbine propulsion assembly of comparable thrust class. The ducted fan assembly **260**, by lessening the proportion of the thrust required to be provided by the unducted fan assembly **104**, may permit a reduction in the overall fan diameter of the unducted fan assembly and thereby provide for installation flexibility and reduced weight.

Operationally, the engine **200** may include a control system that manages the loading of the respective open and ducted fans, as well as potentially the exit area of the variable fan nozzle, to provide different thrust, noise, cooling capacity, and other performance characteristics for various portions of the flight envelope and various operational conditions associated with aircraft operation. For example, in climb mode the ducted fan may operate at maximum pressure ratio there-by maximizing the thrust capability of stream, while in cruise mode, the ducted fan may operate a lower pressure ratio, raising overall efficiency through reliance on thrust from the unducted fan. Nozzle actuation modulates the ducted fan operating line and overall engine fan pressure ratio independent of total engine airflow.

The ducted fan stream flowing through fan duct **266** may include one or more heat exchangers **268** for removing heat from various fluids used in engine operation (such as an air-cooled oil cooler (ACOC), cooled cooling air (CCA), etc.). The heat exchangers **268** may take advantage of the integration into the fan duct **266** with reduced performance penalties (such as fuel efficiency and thrust) compared with traditional ducted fan architectures, due to not impacting the primary source of thrust which is, in this case, the unducted fan stream. Heat exchangers may cool fluids such as gearbox oil, engine sump oil, thermal transport fluids such as supercritical fluids or commercially available single-phase or two-phase fluids (supercritical CO2, EGV, Slither 900, liquid metals, etc.), engine bleed air, etc. Heat exchangers may also be made up of different segments or passages that cool different working fluids, such as an ACOC paired with a fuel cooler. Heat exchangers **268** may be incorporated into a thermal management system which provides for thermal transport via a heat exchange fluid flowing through a network to remove heat from a source and transport it to a heat exchanger.

The fan duct **266** also provides other advantages in terms of reduced nacelle drag, enabling a more aggressive nacelle close-out, improved core stream particle separation, and inclement weather operation. Exhausting the fan duct flow over the engine core cowl **272** aids in energizing the boundary layer and enabling the option of a steeper nacelle close out angle between a maximum dimension of the engine core cowl **272** and the exhaust **256**. The close-out angle is normally limited by air flow separation, but boundary layer energization by air from the fan duct **266** exhausting over the engine core cowl **272** reduces air flow separation. This yields a shorter, lighter structure with less frictional surface drag.

The fan assembly and/or vane assembly can be shrouded or unshrouded (as shown in FIGS. **1** and **2**). Although not shown, an optional annular shroud or duct can be coupled to the vane assembly **210** and located distally from the engine centerline axis **220** relative to the vanes **212**. In addition to the noise reduction benefit, the duct may provide improved vibratory response and structural integrity of the vanes **212** by coupling them into an assembly forming an annular ring or one or more circumferential sectors, i.e., segments forming portions of an annular ring linking two or more of the vanes **212**. The duct may also allow the pitch of the vanes **212** to be varied more easily. For example, FIGS. **3-4**, discussed in more detail below, disclose examples in which both the fan assembly and vane assembly are shrouded.

Although depicted as an unshrouded or open rotor engine in the examples depicted above, it should be appreciated that aspects of the disclosure provided herein may be applied to shrouded or ducted engines, partially ducted engines, aft-fan engines, or other turbomachinery configurations, including those for acro-propulsion systems. Certain aspects of the disclosure may be applicable to turbofan, turboprop, or turboshaft engines.

FIG. **3** is a schematic cross-sectional view of a gas turbine engine in accordance with an example of the present disclosure. More particularly, for the example of FIG. **3**, the gas turbine engine is a high-bypass turbofan engine **300**, referred to herein as "turbofan engine **300**." As shown in FIG. **3**, the turbofan engine **300** defines an axial direction A (extending parallel to a longitudinal axis **302** or centerline provided for reference) and a radial direction R (extending perpendicular to the axial direction A). In general, the turbofan engine **300** includes a fan section **304** and a core engine **306** disposed downstream from the fan section **304**. The turbofan engine **300** also includes a gear assembly or power gear box **336** having a plurality of gears for coupling a gas turbine shaft to a fan shaft. The position of the power gear box **336** is not limited to that as shown in the example of turbofan engine **300**. For example, the position of the power gear box **336** may vary along the axial direction A.

The exemplary core engine **306** depicted generally includes a substantially tubular outer casing **308** that defines an annular inlet **310**. The outer casing **308** encases, in serial flow relationship, a compressor section including a booster or low-pressure (LP) compressor **312** and a high-pressure (HP) compressor **314**; a combustion section **316**; a turbine section including a high-pressure (HP) turbine **318** and a low-pressure (LP) turbine **320**; and a jet exhaust nozzle section **322**. A high-pressure (HP) shaft or spool **324** drivingly connects the HP turbine **318** to the HP compressor **314**. A low-pressure (LP) shaft or spool **326** drivingly connects the LP turbine **320** to the LP compressor **312**. Additionally, the compressor section, combustion section **316**, and turbine section together define at least in part a core air flowpath **327** extending therethrough.

A gear assembly of the present disclosure is compatible with standard fans, variable pitch fans, or other configurations. For the example depicted, the fan section **304** may include a variable pitch fan **328** having a plurality of fan blades **330** coupled to a disk **332** in a spaced-apart manner. As depicted, the fan blades **330** extend outwardly from disk **332** generally along the radial direction R. Each fan blade **330** is rotatable relative to the disk **332** about a pitch axis P by virtue of the fan blades **330** being operatively coupled to a suitable actuation member **334** configured to collectively vary the pitch of the fan blades **330**. The fan blades **330**, disk **332**, and actuation member **334** are together rotatable about the longitudinal axis **302** by LP shaft **326** across a gear assembly **336**. The gear assembly **336** may enable a speed change between a first shaft, e.g., LP shaft **326**, and a second shaft, e.g., LP compressor shaft and/or fan shaft. For example, in some instances, the gear assembly **336** may be disposed in an arrangement between a first shaft and a second shaft such as to reduce an output speed from one shaft to another shaft.

More generally, the gear assembly **336** can be placed anywhere along the axial direction A to decouple the speed of two shafts, whenever it is convenient to do so from a component efficiency point of view, e.g., faster LP turbine and slower fan and LP compressor or faster LP turbine and LP compressor and slower fan.

The gear assembly **336** (which can also be referred to as "a gearbox") can, in some examples, comprise a gear ratio of less than or equal to ten. For example, the gearbox **336** can comprise a gear ratio within a range of 2.0-10.0, 2.0-6.0, 2.0-4.0, 2.0-2.9, 3.0-3.5, 3.2-4.0, 3.25-3.75, 2.3-3.3, 3.0-3.3, etc. In some examples, the gearbox **336** can comprise a gear ratio of 3.5. In some examples, the gearbox **336** can comprise a gear ratio of 3.06. In some examples, the gearbox **336** can comprise a gear ratio of 3.1. In some examples, the gearbox **336** can comprise a gear ratio of 3.2. In some examples, the gearbox **336** can comprise a gear ratio of 3.3.

Referring still to the example of FIG. **3**, the disk **332** is covered by a rotatable front nacelle **338** aerodynamically contoured to promote airflow through the plurality of fan blades **330**. Additionally, the exemplary fan section **304** includes an annular fan casing or outer nacelle **340** that circumferentially surrounds the fan **328** and/or at least a portion of the core engine **306**. The nacelle **340** is, for the example depicted, supported relative to the core engine **306** by a plurality of circumferentially-spaced outlet guide vanes **342**. Additionally, a downstream section **344** of the nacelle **340** extends over an outer portion of the core engine **306** so as to define a bypass airflow passage **346** therebetween.

During operation of the turbofan engine **300**, a volume of air **348** enters the turbofan engine **300** through an associated inlet **350** of the nacelle **340** and/or fan section **304**. As the volume of air **348** passes across the fan blades **330**, a first portion of the air **348**, as indicated by arrows **352**, is directed or routed into the bypass airflow passage **346** and a second portion of the air **348**, as indicated by arrow **354**, is directed or routed into the LP compressor **312**. The ratio between the first portion of air **352** and the second portion of air **354** is commonly known as a bypass ratio. The pressure of the second portion of air **354** is then increased as it is routed through the high-pressure (HP) compressor **314** and into the combustion section **316**, where it is mixed with fuel and burned to provide combustion gases **356**.

The combustion gases **356** are routed through the HP turbine **318** where a portion of thermal and/or kinetic energy from the combustion gases **356** is extracted via sequential stages of HP turbine stator vanes **358** that are coupled to the outer casing **308** and HP turbine rotor blades **360** (e.g., two stage) that are coupled to the HP shaft or spool **324**, thus causing the HP shaft or spool **324** to rotate, thereby supporting operation of the HP compressor **314**. The combustion gases **356** are then routed through the LP turbine **320** where a second portion of thermal and kinetic energy is extracted from the combustion gases **356** via sequential stages of LP turbine stator vanes **362** that are coupled to the outer casing **308** and LP turbine rotor blades **364** (e.g., four stages) that are coupled to the LP shaft or spool **326**, thus causing the LP shaft or spool **326** to rotate, thereby supporting operation of the LP compressor **312** and/or rotation of the fan **328**.

It should be noted that a high-pressure turbine (e.g., the HP turbine **318**) can, in some examples, comprise one or two rotating blade stages and that a low-pressure turbine (e.g., LP turbine **320**) can, in some instances, comprise three, four, five, six, or seven rotating blade stages.

The combustion gases **356** are subsequently routed through the jet exhaust nozzle section **322** of the core engine **306** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **352** is substantially increased as the first portion of air **352** is routed through the bypass airflow passage **346** before it is exhausted from a fan nozzle exhaust section **366** of the turbofan engine **300**, also providing propulsive thrust. The HP turbine **318**, the LP turbine **320**, and the jet exhaust nozzle section **322** at least partially define a hot gas path **368** for routing the combustion gases **356** through the core engine **306**.

FIG. **4** is a cross-sectional schematic illustration of an example of an engine **400** that includes a gear assembly **402** in combination with a ducted fan assembly **404** and a core engine. However, unlike the open rotor configuration of the engine **200** of FIG. **2**, the ducted fan assembly **404** and its fan blades **408** are contained within an annular fan case **480** (which can also be referred to as "a nacelle") and a vane assembly **410** and vanes **412** extend radially between a fan cowl **452** (and/or an engine core cowl **472**) and the inner surface of the fan case **480**. As discussed above, the gear assemblies disclosed herein can provide for increased gear ratios for a fixed gear envelope (e.g., with the same size ring gear), or alternatively, a smaller diameter ring gear may be used to achieve the same gear ratios.

The core engine comprises a compressor section **430**, a combustor section **432**, and a turbine section **434**. The compressor section **430** can include a high-pressure compressor **436** and a booster or a low-pressure compressor **442**. The turbine section **434** can include a high-pressure turbine **438** (e.g., one stage) and a low-pressure turbine **444** (e.g., three stage). The low-pressure compressor **442** is positioned forward of and in flow relationship with the high-pressure compressor **436**. The low-pressure compressor **442** is rotatably coupled with the low-pressure turbine **444** via a low-speed shaft **446** to enable the low-pressure turbine **444** to drive the low-pressure compressor **442** (and a ducted fan **460**). The low-speed shaft **446** is also operably connected to the gear assembly **402** to provide power to the fan assembly **404**. The high-pressure compressor **436** is rotatably coupled with the high-pressure turbine **438** via a high-speed shaft **440** to enable the high-pressure turbine **438** to drive the high-pressure compressor **436**.

It should be noted that a high-pressure turbine (e.g., the high-pressure turbine **438**) can, in some examples, comprise one or two stages and that a low-pressure turbine (e.g., the low-pressure turbine **444**) can, in some instances, comprise three, four, five, or six rotating blade stages.

In some examples, the engine **400** can comprise a pitch change mechanism **482** coupled to the fan assembly **404** and configured to vary the pitch of the fan blades **408**. In certain examples, the pitch change mechanism **482** can be a linear actuated pitch change mechanism.

In some examples, the engine **400** can comprise a variable fan nozzle. Operationally, the engine **400** may include a control system that manages the loading of the fan assembly **404**, as well as potentially the exit area of the variable fan nozzle, to provide different thrust, noise, cooling capacity and other performance characteristics for various portions of the flight envelope and various operational conditions associated with aircraft operation. For example, nozzle actuation modulates the fan operating line and overall engine fan pressure ratio independent of total engine airflow.

The fans disclosed herein (e.g., the fan assemblies **104**, **204**, **304**, and **404**) can comprise various materials. For example, in some instances, a fan can comprise a metal alloy. In some instances, the metal alloy can comprise aluminum, lithium, titanium, and/or other suitable metals for fan blades (e.g., the fan blades **108**, **208**, **330**, and **408**). In some instances, a fan can comprise composite material. In some examples, a fan can comprise a metal alloy core and a composite cover.

The fans disclosed herein (e.g., the fan assemblies **104**, **204**, **304**, and **404**) can comprise various dimensions. For example, a fan can comprise a diameter (as measured at the tip of the leading edge) within a range of 72-120 inches (6-10 feet). In some instances, a fan can comprise a diameter within a range of 84-120 inches (7-10 feet), 80-90 inches, or 84-96 inches (7-8 feet).

The fans disclosed herein comprise a solidity. Solidity is based on average blade chord defined as the blade planform area (surface area on one side of a blade) divided by the blade radial span. The solidity is directly proportional to the number of blades and chord length and inversely proportional to the diameter. For purposes of this disclosure, solidity is equal to the average blade chord (C) times the number of fan blades (N) divided by the product of two (2) times pi (π) times a reference radius (R_ref), which herein is a radius equal to 0.75 times a tip radius of a rotor blade (Rt) (i.e., $C\times N/(2\times\pi\times R\_ref)$). Using this formula, a fan can comprise a solidity from 0.5 to 1.0, or more particularly from 0.6 to 1.0. In other examples, a fan can comprise a solidity from 1.1 to 1.5, or 1.1-1.3 in certain examples. In still other examples, enhanced performance can be observed when the solidity is greater than or equal to 0.8 and less than or equal to 2, greater than or equal to 0.8 and less than or equal to 1.5, greater than or equal to 1 and less than or equal to 2, or greater than or equal to 1.25 and less than or equal to 1.75.

As mentioned above, rising fuel prices, depleting natural resources, and regulatory constraints place increasing demands on turbomachinery engines. As such, turbomachinery engines with improved efficiency and performance are desired. Designing turbomachinery engines, however, is complex, time consuming, and expensive. There are many engine components and parameters to consider (each of various weight), and many are of the components and parameters are interdependent. Therefore, changing one component or one parameter can often create cascading effects requiring one or more other parameters or components to be reconfigured.

Various turbomachinery engines and gear assemblies are disclosed herein. The disclosed turbomachinery engines have improved efficiency and/or performance than typical turbomachinery engines.

The disclosed turbomachinery engines comprise a gearbox and a turbine (e.g., a low-pressure turbine) coupled to the gearbox. The disclosed turbomachinery engines are characterized or defined by one or more parameters of a turbine (e.g., the low-pressure turbine). These turbine parameters include: an area ratio and/or an area-EGT ratio.

After numerous engine designs, the inventors found unexpectedly that engines comprising the area ratio ranges and/or the area-EGT ratio ranges disclosed herein provide a turbomachine engine with improved performance and efficiency, as discussed further below.

The low-pressure turbines disclosed herein comprise 3-4 rotating stages and an area ratio within a range of 2.0-5.1. The inventors discovered that low-pressure turbines comprising 3-4 stages and an area ratio within a range of 2.0-5.1 are particularly advantageous.

Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio equals the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine.

The low-pressure turbines disclosed herein additionally comprise an area-EGT ratio within a range of 1.06-1.6, 1.2-1.6, 1.2-1.3, 1.25-1.35, 1.3-1.6. The area-EGT ratio= $(\text{area ratio})^{(1/(stages-1))}/(\text{EGT}/1000)$. Each rotating stage comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio is the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, wherein the stages is the number of rotating stages of the low-pressure turbine.

The term "redline exhaust gas temperature" (referred to herein as "redline EGT") refers to a maximum permitted takeoff temperature documented in a Federal Aviation Administration ("FAA") type certificate data sheet. For example, in certain examples, the term redline EGT may refer to a maximum takeoff temperature of an airflow after a first stage stator downstream of an HP turbine of an engine that the engine is rated to withstand. In other examples, the term redline EGT refers to a maximum temperature of an airflow after the first stator downstream of the last stage of rotor blades of the HP turbine and into the first of the plurality of LP turbine rotor blades **210**. The term redline EGT is sometimes also referred to as an indicated turbine temperature.

The term "redline operating condition" refers to the maximum permissible engine rotor operating speed documented in a FFA type certificate data sheet. In certain examples, the FFA type certificate may provide a redline operating condition as the maximum permissible engine rotor speed for the low-pressure rotor (N1) and/or high-pressure rotor (N2) stated in revolutions per minute (rpm).

In some examples, a turbomachinery engine includes a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine includes 3-4 rotating stages. Each rotating stage of the low-pressure turbine includes an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The low-pressure turbine includes an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and the area ratio is within a range of 2.0-5.1. The gearbox includes an input and an output. The input of the gearbox is coupled to the low-pressure turbine and includes a first rotational speed, and the output of the gearbox is coupled to the fan assembly and has a second rotational speed.

In some examples, the area ratio of the low-pressure turbine is within a range of 2.2-2.6.

In some examples, the area ratio of the low-pressure turbine is within a range of 2.0-3.5.

In some instances, the low-pressure turbine includes exactly three rotating stages and/or the area ratio of the low-pressure turbine is within a range of 2.2-3.0.

In some instances, the low-pressure turbine includes exactly four rotating stages, and/or the area ratio of the low-pressure turbine is within a range of 2.0-5.1, or 2.3-3.3, or 2.3-2.6, or 2.3-2.9.

In some examples, a turbomachinery engine includes a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine comprising 3-4 rotating stages. Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and the area ratio is within a range of 2.0-4.6 (or 2.0-3.5 or 2.25-2.6). The gearbox including an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, the output of the gearbox is coupled to the fan assembly and has a second rotational speed, and a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.5.

In some examples, a turbomachinery engine comprising a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly including a plurality of fan blades. The low-pressure turbine comprises 3-4 rotating stages and an area-EGT ratio within a range of 1.05-1.6. The

$$\text{area-}EGT\text{ ratio} = \frac{(\text{area ratio})^{(1/(LPT\ stages-1))}}{(EGT/1000)}.$$

Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio is the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine. The LPT stages is the number of rotating stages of the low-pressure turbine. The EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition. The gearbox including an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, and the output of the gearbox is coupled to the fan assembly and has a second rotational speed.

In some examples, the area-EGT ratio is within a range of 1.05-1.58.

In some examples, the area-EGT ratio is within a range of 1.05-1.53.

In some examples, the area-EGT ratio is within a range of 1.05-1.30.

In some examples, the area-EGT ratio is within a range of 1.20-1.30.

In some examples, the area-EGT ratio is within a range of 1.3-1.6.

It should be noted that there are some engines described herein (e.g., FIGS. **13-16**) that have an area ratio and/or area-EGT ratio outside of the ranges of 2.0-5.1 and 1.05-1.6, respectively. These engines are provided merely for purposes of comparison.

FIG. **5**A depicts a portion of a three-stage low-pressure turbine **500**, according to one example of the disclosed technology. The low-pressure turbine (LPT) **500** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The LPT **500** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **500** comprises three rotating blade stages **502***a*, **502***b*, and **502***c* and two stationary vane stages **504***a* and **504***b*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **502**" or simply "the blades **502**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **504**" or simply "the vanes **504**."

The blades **502** and the vanes **504** are disposed within a duct **506**, which guides the fluid flow through the LPT **500**.

Each rotating blade stage **502** of the LPT **500** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

For example, the forward-most rotating blade stage (which can also be referred to as "the first stage") **502***a* of the LPT **500** comprises an annular exit area **508***a*, as depicted in FIG. **5**B. The annular exit area **508***a* is defined by the tip radius R_{tip1} and hub radius R_{hub1}. R_{tip1} is the tip radius of the trailing edge of any blade of the first stage **502***a* (or a nominal tip radius of the trailing edges of the blades of the first stage **502***a*), and R_{hub1} is the hub radius of the blade (or a nominal hub radius of the blades of the first stage) at the axial location aligned with the tip radius R_{tip1}.

In some examples, the annular exit area of the first stage **502***a* can be within a range of 155-380 in^2 or within a range of 155-372 in^2. In particular examples, the annular exit area can be within a range of 280-380 in^2 or within a range of 285-372 in^2. In the depicted example, the annular exit area

508a of the first stage **502**a is about 327 in^2. Additional examples of annular exit areas for the first stage of a three-stage low-pressure turbine are provided in the table depicted in FIG. **9**.

As another example, the second stage **502**b of the LPT **500** comprises an annular exit area. The annular exit area of the second stage **502**b is defined by the tip radius R_{tip2} and hub radius R_{hub2}. R_{tip2} is the tip radius of the trailing edge of any blade of the second stage **502**b (or a nominal tip radius of the trailing edges of the blades of the second stage **502**b), and R_{hub2} is the hub radius of the blade (or a nominal hub radius of the blades of the second stage **502**b) at the axial location aligned with the tip radius R_{tip2}.

In some examples, the annular exit area of the second stage **502**b can be within a range of 230-750 in^2 or within a range of 250-700 in^2. In particular examples, the annular exit area of the second stage **502**b can be within a range of 450-750 in^2 or within a range of 462-699 in^2. In the depicted example, the annular exit area of the second stage **502**b is about 526 in^2. Additional examples of annular exit areas for the second stage of a three-stage low-pressure turbine are provided in the table depicted in FIG. **9**.

As another example, the aft-most stage (which can also be referred to as "the third stage") **502**c of the LPT **500** comprises an annular exit area. The annular exit area of the third stage **502**c is defined by the tip radius R_{tip3} and hub radius R_{hub3}. R_{tip3} is the tip radius of the trailing edge of any blade of the third stage **502**c (or a nominal tip radius of the trailing edges of the blades of the third stage **502**c), and R_{hub3} is the hub radius of the blade (or a nominal hub radius of the blades of the third stage **502**c) at the axial location aligned with the tip radius R_{tip3}.

In some examples, the annular exit area of the third stage **502**c can be within a range of 350-1050 in^2 or within a range of 379-1027 in^2. In particular examples, the annular exit area of the third stage **502**c can be within a range of 600-1050 in^2 or within a range of 639-1027 in^2. In the depicted example, the annular exit area of the third stage **502**c is about 725 in^2. Additional examples of annular exit areas for the third stage of a low-pressure turbine are provided in the table depicted in FIG. **9**.

The LPT **500** comprises an area ratio (which can also be referred to as "an exit area ratio") within a range of 2.0-5.1, within a range of 2.0-3.0, within a range of 2.2-2.91, and specifically about 2.2. The area ratio equals the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine. For example, for the LPT **500**, the area ratio equals the annular exit area of the third stage **502**c divided by the annular exit area **508**a of the first stage **502**a.

In addition to having an area ratio within a range of 2.0-5.1, the LPT **500** can comprise an area-exhaust gas temperature (EGT) ratio, referred to herein as area-EGT ratio, within a range of 1.05-1.6, within a range of 1.05-1.3, or within a range of 1.38-1.58, and specifically about 1.38. The area-EGT ratio is defined according to Expression (1):

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{(1/(LPT\ stages-1))}}{(EGT/1000)} \quad (1)$$

where the area ratio is as defined above, LPT stages is the number of rotating blade stages of LPT, and EGT is an exhaust gas temperature of the LPT measured in degrees Celsius at an inlet of the LPT at a redline operating condition.

In some examples, the number of LPT stages is 3, 4, or 5. For example, the LPT **500** includes exactly three stages. The table of FIG. **9** provides additional exemplary engines comprising exactly three LPT stages.

In some examples, EGT is within a range of 1060-1180 degrees Celsius measured at the inlet of the LPT at the redline operating condition. For example, the EGT of the LPT **500** is about 1083 degrees Celsius at the redline operating condition. As used herein the inlet of the LPT is defined by the turbine vane frame (TVF). The EGT can be measured at any axial location aligned with the TVF, i.e., from the leading edge to the trailing edge of the TVF. Thus, with respect to the LPT **500**, the inlet of the LPT **500** for purposes of measuring EGT is any axial location aligned with a TVF **510**.

FIG. **6** depicts a portion of a low-pressure turbine **600**, according to one example of the disclosed technology. The low-pressure turbine **600** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**), and particularly Engine **05** depicted in the table of FIG. **9**. The LPT **600** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **600** comprises three rotating blade stages **602**a, **602**b, and **602**c and two stationary vane stages **604**a and **604**b. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **602**" or simply "the blades **602**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **604**" or simply "the vanes **604**."

The blades **602** and the vanes **604** are disposed within a duct **606** aft of a TVF **610**, which guides the fluid flow through the LPT **600**.

Each rotating blade stage **602** of the LPT **600** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius.

For example, the forward-most rotating blade stage (which can also be referred to as "the first stage") **602**a of the LPT **600** comprises an annular exit area. The annular exit area is defined by the tip radius R_{tip1} and hub radius R_{hub1}. R_{tip1} is the tip radius of the trailing edge of any blade of the first stage **602**a (or a nominal tip radius of the trailing edges of the blades of the first stage **602**a), and R_{hub1} is the hub radius of the blade (or a nominal hub radius of the blades of the first stage) at the axial location aligned with the tip radius R_{tip1}.

In some examples, the annular exit area of the first stage **602**a can be within a range of 155-380 in^2. In particular examples, the annular exit area can be within a range of 280-380 in^2 or within a range of 285-372 in^2. In the depicted example, the annular exit area of the first stage **602**a is about 327 in^2.

As another example, the second stage **602**b of the LPT **600** comprises an annular exit area. The annular exit area of the second stage **602**b is defined by the tip radius R_{tip2} and hub radius R_{hub2}. R_{tip2} is the tip radius of the trailing edge of any blade of the second stage **602**b (or a nominal tip radius of the trailing edges of the blades of the second stage **602**b), and R_{hub2} is the hub radius of the blade (or a nominal hub radius of the blades of the second stage **602**b) at the axial location aligned with the tip radius R_{tip2}.

In some examples, the annular exit area of the second stage **602***b* can be within a range of 230-750 in^2 or within a range of 250-700 in^2. In particular examples, the annular exit area of the second stage **602***b* can be within a range of 450-750 in^2 or within a range of 462-699 in^2. In the depicted example, the annular exit area of the second stage **602***b* is about 577 in^2.

As another example, the aft-most stage (which can also be referred to as "the third stage") **602***c* of the LPT **600** comprises an annular exit area. The annular exit area of the third stage **602***c* is defined by the tip radius R_{tip3} and hub radius R_{hub3}. R_{tip3} is the tip radius of the trailing edge of any blade of the third stage **602***c* (or a nominal tip radius of the trailing edges of the blades of the third stage **602***c*), and R_{hub3} is the hub radius of the blade (or a nominal hub radius of the blades of the third stage **602***c*) at the axial location aligned with the tip radius R_{tip3}.

In some examples, the annular exit area of the third stage **602***c* can be within a range of 350-1050 in^2 or within a range of 379-1027 in^2. In particular examples, the annular exit area of the third stage **602***c* can be within a range of 700-1050 in^2 or within a range of 639-1027 in^2. In the depicted example, the annular exit area of the third stage **602***c* is about 827 in^2.

The LPT **600** comprises an area ratio (which can also be referred to as "an exit area ratio") within a range of 2.0-5.1, within a range of 2.0-3.0, within a range of 2.2-2.9, and specifically about 2.53. For example, for the LPT **600**, the area ratio equals the annular exit area of the third stage **602***c* divided by the annular exit area of the first stage **602***a.*

The LPT **600** can also comprise an area-EGT ratio within a range of 1 . . . 05-1.6, within a range of 1.35-1.58, and specifically about 1.47.

In some examples, the EGT of the LPT **600** is within a range of 1060-1180 degrees Celsius measured at the inlet of the LPT at the redline operating condition. For example, the LPT **600** comprises an EGT of about 1083 degrees Celsius at the redline operating condition.

FIG. **7** depicts a portion of a low-pressure turbine **700**, according to one example of the disclosed technology. The low-pressure turbine **700** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**), and particularly Engine **07** depicted in the table of FIG. **9**. The LPT **700** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **700** comprises three rotating blade stages **702***a*, **702***b*, and **702***c* and two stationary vane stages **704***a* and **704***b*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **702**" or simply "the blades **702**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **704**" or simply "the vanes **704**."

The blades **702** and the vanes **704** are disposed within a duct **706** aft of a TVF **710**, which guides the fluid flow through the LPT **700**.

Each rotating blade stage **702** of the LPT **700** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

For example, the forward-most rotating blade stage (which can also be referred to as "the first stage") **702***a* of the LPT **700** comprises an annular exit area. The annular exit area is defined by the tip radius R_{tip1} and hub radius R_{hub1}. R_{tip1} is the tip radius of the trailing edge of any blade of the first stage **702***a* (or a nominal tip radius of the trailing edges of the blades of the first stage **702***a*), and R_{hub1} is the hub radius of the blade (or a nominal hub radius of the blades of the first stage) at the axial location aligned with the tip radius R_{tip1}.

In some examples, the annular exit area of the first stage **702***a* can be within a range of 155-380 in^2. In particular examples, the annular exit area can be within a range of 280-380 in^2. In the depicted example, the annular exit area of the first stage **702***a* is about 372 in^2.

As another example, the second stage **702***b* of the LPT **700** comprises an annular exit area. The annular exit area of the second stage **702***b* is defined by the tip radius R_{tip2} and hub radius R_{hub2}. R_{tip2} is the tip radius of the trailing edge of any blade of the second stage **702***b* (or a nominal tip radius of the trailing edges of the blades of the second stage **702***b*), and R_{hub2} is the hub radius of the blade (or a nominal hub radius of the blades of the second stage **702***b*) at the axial location aligned with the tip radius R_{tip2}.

In some examples, the annular exit area of the second stage **702***b* can be within a range of 250-710 in^2. In particular examples, the annular exit area of the second stage **702***b* can be within a range of 450-700 in^2. In the depicted example, the annular exit area of the second stage **702***b* is about 700 in^2.

As another example, the aft-most stage (which can also be referred to as "the third stage") **702***c* of the LPT **700** comprises an annular exit area. The annular exit area of the third stage **702***c* is defined by the tip radius R_{tip3} and hub radius R_{hub3}. R_{tip3} is the tip radius of the trailing edge of any blade of the third stage **702***c* (or a nominal tip radius of the trailing edges of the blades of the third stage **702***c*), and R_{hub3} is the hub radius of the blade (or a nominal hub radius of the blades of the third stage **702***c*) at the axial location aligned with the tip radius R_{tip3}.

In some examples, the annular exit area of the third stage **702***c* can be within a range of 350-1100 in^2. In particular examples, the annular exit area of the third stage **702***c* can be within a range of 380-1050 in^2 or within a range of 639-1027 in^2. In the depicted example, the annular exit area of the third stage **702***c* is about 1027 in^2.

The LPT **700** comprises an area ratio (which can also be referred to as "an exit area ratio") within a range of 2.0-5.1, within a range of 2.22-2.91, and specifically about 2.76. For example, for the LPT **700**, the area ratio equals the annular exit area of the third stage **702***c* divided by the annular exit area of the first stage **702***a.*

The LPT **700** can, additionally or alternatively to the area ratio within a range of 2.0-5.1, comprise an area-EGT ratio within a range of 1.05-1.6, within a range of 1.35-1.55, and specifically 1.53.

In some examples, EGT of the LPT **700** is within a range of 1060-1180 degrees Celsius measured at the inlet of the LPT at the redline operating condition. For example, the LPT **700** comprises an EGT of about 1083 degrees Celsius at the redline operating condition.

FIG. **8** depicts a portion of a low-pressure turbine **800**, according to one example of the disclosed technology. The low-pressure turbine **800** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**), and particularly Engine **09** depicted in the table of FIG. **9**. The LPT **800** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **800** comprises three rotating blade stages **802***a*, **802***b*, and **802***c* and two stationary vane stages **804***a* and **804***b*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **802**" or simply "the blades **802**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **804**" or simply "the vanes **804**."

The blades **802** and the vanes **804** are disposed within a duct **806** aft of a TVF **810**, which guides the fluid flow through the LPT **800**.

Each rotating blade stage **802** of the LPT **800** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

The LPT **800** comprises three rotating blade stages, a redline EGT of 1067 degrees Celsius, a first stage exit area of 293.2 in^2, a second stage exit area of 476 in^2, a third stage exit area of 764.9 in^2, an area ratio of 2.61, an area-EGT ratio of 1.51, a first stage AN^2 value of 30, and a third stage AN^2 value of 80. AN^2 the product of A and N^2, where A is the annular exit area of a particular rotating stage of the low-pressure turbine measured in square inches, N is the rotational speed of the low-pressure turbine measured in revolutions per minute at a redline operating condition, and the product of AN^2 is divided by 10^9.

FIG. **9** provides additional information about the LPT **800** (see Engine **09**).

FIG. **9** provides a table with several additional examples of turbomachinery engines comprising three rotating blade stages, a LPT with an area ratio within a range of 2.0-5.1 (particularly 2.22-2.91) and an area-EGT ratio within a range of 1.05-1.6 (particularly 1.38-1.58). The engines disclosed in FIG. **9** comprise a gear ratio of 2-9 or 2.95-8.33. The EGT at a redline operating condition for the engines of FIG. **9** is within a range of 1060-1175 degrees Celsius or within a range of 1067-1083 degrees Celsius. The exit area of stage **1** of the disclosed engines is within a range of 155-380 in^2 or within a range of 285.0-372.4 in^2. The exit area of stage **2** of the engines of FIG. **9** is within a range of 230-750 in^2 or within a range of 461.8-699.5 in^2. The exit area of stage **3** of the engines of FIG. **9** is within a range of 350-1050 in^2 or within a range of 638.5-1027.82 in^2. The area ratio of the engines disclosed in FIG. **9** is within a range of 2.0-5.1 or within a range of 2.22-2.91. The area-EGT ratio is within a range of 1.05-1.6 or within a range of 1.38-1.58. The engines disclosed in FIG. **9** comprise a first stage AN^2 value within a range of 9-36 or within a range of 30-36 at a redline operating condition. The engines disclosed in FIG. **9** comprise a third stage (exit) AN^2 value within a range of 44-105 or within a range of 78-105 at a redline operating condition.

FIGS. **10-12** provide examples of low-pressure turbines comprising four rotating blade stages. The disclosed LPTs comprise an area ratio within a range of 2.0-5.1 (particularly 2.05-5.0, 2.0-3.5) and an area-EGT ratio within a range of 1.05-1.6 (particularly 1.09-1.58).

FIG. **10** depicts a portion of a four-stage low-pressure turbine **900**, according to one example of the disclosed technology. The low-pressure turbine **900** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The LPT **900** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **900** comprises four rotating blade stages **902***a*, **902***b*, **902***c*, and **902***d* and three stationary vane stages **904***a*, **904***b*, and **904***c*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **902**" or simply "the blades **902**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **904**" or simply "the vanes **904**."

It was found that a four-stage high speed low-pressure turbine (LPT) provides an improved geared engine configuration that best accommodates advances in propulsion and provides the balance between engine weight, engine size, and the efficient conversion of kinetic energy into mechanical power for driving a fan and a low-pressure compressor (which can also be referred to as a "booster"). Additional work will be needed as overall pressure ratios (OPR) are increased to improve engine performance while maintaining the pressure ratio of a known high-pressure compressor (HPC) design. For future engines as temperatures increase for improved thermal efficiency, the core gets smaller, which increases the loading on the low-pressure turbine. This will require an increased stage count. Utilizing an existing or scaled HPC design will minimize development cost by using the same or slightly improved characteristics on the HPC of an in-service engine. The four-stage LPT can balance loading from the additional work to maximize efficiency while limiting weight addition and size increases. A higher LPT stage count (e.g., 5 or higher) may in some instances also achieve a target loading but at the expense of a longer, heavier, and more expensive LPT design. A three-stage design, can, in some instances, require a max radius increase and/or increased rotor speed.

The desired area ratio range for a four-stage high speed LPT, as identified by the inventors, avoids the drawbacks for a design that falls outside of the area ratio range of 2.0-5.1. A four-stage LPT design with an area ratio higher than 5.1 can require (1) an increased flowpath slope resulting in additional secondary flow losses and an increased risk of separation, and/or (2) an increased length resulting in larger profile losses, weight, and installation penalties. A four-stage LPT design with an area ratio below 2.0 can result in higher than desired Mach numbers through the LPT. Undesirably high Mach numbers can produce additional losses in the LPT and/or other downstream components. FIG. **13** provides several examples of four-stage LPT designs that would produce a less desired outcome (for one or more of the above reasons) than the exemplary four-stage LPT designs provided in FIG. **12**. For example, Engine **33** (FIG. **13**) is generally similar to Engine **29** (FIG. **12**), but Engine **33** has an area ratio less than 2.0 (i.e., 1.8). Accordingly, Engine **33** will have higher than desired Mach numbers through the LPT resulting in additional losses in the LPT and/or downstream components. As another example, Engine **38** (FIG. **13**) is generally similar to Engine **30** (FIG. **12**), but Engine **38** has an area ratio higher than 5.1 (i.e., 6.75). As such, Engine **38** has an undesirably high flowpath slope resulting in additional secondary flow losses and an increased risk of separation.

In some instances, a LPT with an area ratio of 2.0-3.5 can be particularly advantageous. Configuring an LPT with an area ratio within this range can, for example, result in optimum fuel burn for the engine by balancing profile and secondary loss plus weight and installation effects.

The blades **902** and the vanes **904** are disposed within a duct **906** aft of a TVF **910**, which guides the fluid flow through the LPT **900**.

Each rotating blade stage **902** of the LPT **900** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

The LPT **900** comprises four rotating blade stages, a redline EGT of 1080 degrees Celsius, a first stage exit area of 299.2 in^2, a second stage exit area of 442.1 in^2, a third stage exit area of 618.3 in^2, a fourth stage exit area of 998.1 in^2, an area ratio of 3.34, an area-EGT ratio of 1.38, a first stage AN^2 value of 13, and a fourth stage AN^2 value of 44.

FIG. **11** depicts a portion of a four-stage low-pressure turbine **1000**, according to one example of the disclosed technology. The low-pressure turbine **1000** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The LPT **1000** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **1000** comprises four rotating blade stages **1002***a*, **1002***b*, **1002***c*, and **1002***d* and three stationary vane stages **1004***a*, **1004***b*, and **1004***c*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **1002**" or simply "the blades **1002**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **1004**" or simply "the vanes **1004**."

The blades **1002** and the vanes **1004** are disposed within a duct **1006** aft of a TVF **1010**, which guides the fluid flow through the LPT **1000**.

Each rotating blade stage **1002** of the LPT **1000** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

The LPT **1000** comprises four rotating blade stages, a redline EGT of 1175 degrees Celsius, a first stage exit area of 222.4 in^2, a second stage exit area of 350.7 in^2, a third stage exit area of 612.1 in^2, a fourth stage exit area of 907.7 in^2, an area ratio of 4.08, an area-EGT ratio of 1.36, a first stage AN^2 value of 20, and a fourth stage AN^2 value of 80.

FIG. **12** provides a table with several additional examples of turbomachinery engines comprising four rotating blade stages, a LPT with an area ratio within a range of 2.0-3.5 and an area-EGT ratio within a range of 1.05-1.6. The engines disclosed in FIG. **12** comprise a gear ratio of 2.0-9.0 or 2.6-8.70. The EGT at a redline operating condition for the engines of FIG. **12** is within a range of 1067-1175 degrees Celsius. The exit area of stage **1** of the disclosed engines is within a range of 155-380 in^2 or within a range of 171.9-299.2 in^2. The exit area of stage **2** of the engines of FIG. **12** is within a range of 230-750 in^2 or within a range of 250-579.47 in^2. The exit area of stage **3** of the engines of FIG. **12** is within a range of 330-1050 in^2 or within a range of 339-944 in^2. The exit area of stage **4** of the engines of FIG. **12** is within a range of 420-1400 in^2 or within a range of 429-1309 in^2. The area ratio of the engines disclosed in FIG. **12** is within a range of 2.0-5.1 or within a range of 2.0-3.06 or within a range of 3.06-5.09 (3.1-5.1). The area-EGT ratio is within a range of 1.05-1.6 or within a range of 1.05-1.39. The engines disclosed in FIG. **12** comprise a first stage AN^2 value within a range of 13-22 at a redline operating condition. The engines disclosed in FIG. **12** comprise a fourth stage (exit) AN^2 value within a range of 29-105 at a redline operating condition.

FIGS. **14-16** provide examples of low-pressure turbines comprises five rotating blade stages. The disclosed LPTs comprise an area ratio within a range of greater than 5.1, which have the drawbacks discussed above.

FIG. **14** depicts a portion of a five-stage low-pressure turbine **1100**, according to one example of the disclosed technology. The low-pressure turbine **1100** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**), and particularly Engine **41** depicted in the table of FIG. **16**. The LPT **1100** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **1100** comprises five rotating blade stages **1102***a*, **1102***b*, **1102***c*, **1102***d*, and **1102***e* and four stationary vane stages **1104***a*, **1104***b*, **1104***c*, and **1104***d*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **1102**" or simply "the blades **1102**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **1104**" or simply "the vanes **1104**."

The blades **1102** and the vanes **1104** are disposed within a duct **1106** aft of a TVF **1110**, which guides the fluid flow through the LPT **1100**.

Each rotating blade stage **1102** of the LPT **1100** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

The LPT **1100** comprises five rotating blade stages, a redline EGT of 1175 degrees Celsius, a first stage exit area of 212.1 in^2, a second stage exit area of 341.6 in^2, a third stage exit area of 524.5 in^2, a fourth stage exit area of 875.0 in^2, a fifth stage exit area of 1212.0 in^2, an area ratio of 5.72, an area-EGT ratio of 1.32, a first stage AN^2 value of 15, and a fifth stage AN^2 value of 84. FIG. **16** provides additional information about the LPT **1100** (see Engine **41**).

FIG. **15** depicts a portion of a five-stage low-pressure turbine **1200**, according to one example of the disclosed technology. The low-pressure turbine **1200** can be used, for example, with any of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**), and particularly Engine **44** depicted in the table of FIG. **16**. The LPT **1200** comprises a plurality of rotating blade stages and a plurality of stationary vane stages. In particular, the depicted portion of the LPT **1200** comprises five rotating blade stages **1202***a*, **1202***b*, **1202***c*, **1202***d*, and **1202***e* and four stationary vane stages **1204***a*, **1204***b*, **1204***c*, and **1204***d*. The rotating blade stages are referred to herein generically or collectively as "a/the rotating blade stage(s) **1202**" or simply "the blades **1202**," and the stationary vane stages are referred to herein generically or collectively as "a/the stationary vane stage(s) **1204**" or simply "the vanes **1204**."

The blades **1202** and the vanes **1204** are disposed within a duct **1206** aft of a TVF **1210**, which guides the fluid flow through the LPT **1200**.

Each rotating blade stage **1202** of the LPT **1200** comprises an annular exit area defined by a tip radius of a trailing edge of any blade of the rotating stage (or a nominal tip radius of the stage) and a hub radius of the blade of the rotating stage (or a nominal hub radius of the stage) at the axial location aligned with the tip radius. With respect shrouded turbine blades, the tip radius is the radius at the tip of the blade portion, excluding the shroud portion.

The LPT **1200** comprises five rotating blade stages, a redline EGT of 1175 degrees Celsius, a first stage exit area of 232.6 in^2, a second stage exit area of 326.9 in^2, a third stage exit area of 527.7 in^2, a fourth stage exit area of 895.0 in^2, a fifth stage exit area of 1279.3 in^2, an area ratio of 5.5, an area-EGT ratio of 1.30, a first stage AN^2 value of 14, and a fifth stage AN^2 value of 76. FIG. **16** provides additional information about the LPT **1200** (see Engine **44**).

FIG. **16** provides a table with several additional examples of turbomachinery engines comprising five rotating blade stages, a LPT with an area ratio within a range of 5.4-6.5 and an area-EGT ratio within a range of 1.3-1.6. Due to their area ratio exceeding 5.1, these engines would not perform as well as engines comprising a LPT with an area ratio of 2.0-5.1. The engines disclosed in FIG. **16** comprise a gear ratio of 2.0-9.0 or 6.96-7.56. The EGT at a redline operating condition for the engines of FIG. **15** is within a range of 1060-1175 degrees Celsius, and particularly 1175 degrees Celsius. The exit area of stage **1** of the disclosed engines is within a range of 155-380 in^2 or within a range of 155.4-232.6 in^2. The exit area of stage **2** of the engines of FIG. **16** is within a range of 230-750 in^2 or within a range of 250.2-341.6 in^2. The exit area of stage **3** of the engines of FIG. **16** is within a range of 350-1050 in^2 or within a range of 379.3-527.7 in^2. The exit area of stage **4** of the engines of FIG. **16** is within a range of 630-1200 in^2 or within a range of 563.1-895 in^2. The exit area of stage **5** of the engines of FIG. **16** is within a range of 800-1300 in^2, within a range of 851-1280 in^2, or within a range of 851.4-1279.3 in^2. The area ratio of the engines disclosed in FIG. **16** is within a range of 5.4-6.5 or within a range of 5.48-6.43. The area-EGT ratio is within a range of 1.3-1.6 or within a range of 1.30-1.36. The engines disclosed in FIG. **16** comprise a first stage AN^2 value within a range of 9-36 or within a range of 9-15 at a redline operating condition. The engines disclosed in FIG. **16** comprise a fifth stage (exit) AN^2 value within a range of 44-104 or within a range of 59-84 at a redline operating condition.

The 3-4 stage low-pressure turbines disclosed herein comprising an area ratio within a range of 2.0-5.1 and an area-EGT ratio within a range of 1.05-1.6 provides one or more advantages over conventional low-pressure turbines. In some examples, the disclosed 3-4 stage LPTs have up to +1.3% (e.g., +0.1% to +1.3%) LPT efficiency compared to conventional LPTs. In some examples, the disclosed LPTs enable reduced LPT stage count or reduced tip speeds, which provides weight and/or cost reduction, without an efficiency penalty. In some examples, the disclosed LPTs enable higher BPR engines without adding LPT stages. In some examples, the disclosed LPTs reduce turbine rear frame (TRF) loss by up to 0.3% dP/P_1 due to the reduced LPT exit Mach number. As used herein, "dP" is the change in fluid pressure across the TRF, and "P_1" is the fluid pressure prior to the TRF. Stated another way, dP/P_1 equals the fluid pressure after the TRF (P_2) minus the fluid pressure prior to the TRF (P_1) divided by P_1. Thus, dP/P_1 is the relative change of the fluid pressure across the TRF. In at least some instances, the LPT exit Mach number of the LPTs disclosed herein can be <0.48.

FIG. **17** schematically depicts a gearbox **1300** that can be used with the engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The gearbox **1300** comprises a two-stage star configuration.

The first stage of the gearbox **1300** includes a first-stage sun gear **1302**, a first-stage carrier **1304** housing a plurality of first-stage star gears, and a first-stage ring gear **1306**. The first-stage sun gear **1302** can be coupled to a low-speed shaft **1308**, which in turn is coupled to a low-pressure turbine. The first-stage sun gear **1302** can mesh with the plurality of first-stage star gears, which mesh with the first-stage ring gear **1306**. The first-stage carrier **1304** can be fixed from rotation by a support member **1310**.

The second stage of the gearbox **1300** includes a second-stage sun gear **1312**, a second-stage carrier **1314** housing a plurality of second-stage star gears, and a second-stage ring gear **1316**. The second-stage sun gear **1312** can be coupled to a shaft **1318** which in turn is coupled to the first-stage ring gear **1306**. The second-stage carrier **1314** can be fixed from rotation by a support member **1320**. The second-stage ring gear **1316** can be coupled to a fan shaft **1322**.

In some examples, each stage of the gearbox **1300** can comprise five star gears. In other examples, the gearbox **1300** can comprise fewer or more than five star gears in each stage. In some examples, the first-stage carrier **1304** can comprise a different number of star gears than the second-stage carrier **1314**. For example, the first-stage carrier **1304** can comprise five star gears, and the second-stage carrier **1314** can comprise three star gears, or vice versa.

FIG. **18** schematically depicts a gearbox **1400** that can be used with the engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The gearbox **1400** comprises a single-stage star configuration. The gearbox **1400** includes a sun gear **1402**, a carrier **1404** housing a plurality of star gears (e.g., 3-5 star gears), and a ring gear **1406**. The sun gear **1402** can mesh with the plurality of star gears, and the plurality of star gears can mesh with the ring gear **1406**. The sun gear **1402** can be coupled to a low-speed shaft **1408**, which in turn is coupled to the low-pressure turbine. The carrier **1404** can be fixed from rotation by a support member **1410**. The ring gear **1406** can be coupled to a fan shaft **1412**.

FIG. **19** schematically depicts a gearbox **1500** that can be used with the engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**). The gearbox **1500** comprises a single-stage star configuration. The gearbox **1500** includes a sun gear **1502**, a carrier **1504** housing a plurality of star gears (e.g., 3-5 star gears), and a ring gear **1506**. The sun gear **1502** can mesh with the plurality of star gears, and the star gears can mesh with the ring gear **1506**. The sun gear **1502** can be coupled to a low-speed shaft **1508**, which in turn is coupled to the low-pressure turbine. The carrier **1504** can be fixed from rotation by a support member **1510**. The ring gear **1506** can be coupled to a fan shaft **1512**.

FIG. **20** depicts a gearbox **1600** that can be used, for example, with the engines disclosed herein (e.g., the engines **100**, **200**, **400**). The gearbox **1600** is configured as a compound star gearbox. The gearbox **1600** comprises a sun gear **1602** and a star carrier **1604**, which includes a plurality of compound star gears having one or more first portions **1606** and one or more second portions **1608**. The gearbox **1600** further comprises a ring gear **1610**. The sun gear **1602** can also mesh with the first portions **1606** of the plurality of compound star gears. The star carrier can be fixed from rotation via a support member **1614**. The second portions **1608** of the plurality of compound star gears can mesh with the ring gear **1610**. The sun gear **1602** can be coupled to a low-pressure turbine via the turbine shaft **1612**. The ring gear **1610** can be coupled to a fan shaft **1616**.

The gear assemblies shown and described herein can be used with any suitable engine. For example, although FIG. **4** shows an optional ducted fan and optional fan duct (similar to that shown in FIG. **2**), it should be understood that such gear assemblies can be used with other ducted turbofan engines (e.g., the engine **300**) and/or other open rotor engines that do not have one or more of such structures.

Configurations of the gear assemblies depicted and described herein may provide for gear ratios and arrangements that fit within the L/D_{core} constraints of the disclosed engines. In certain examples, the gear assemblies depicted and described in regard to FIGS. **17**-**20** allow for gear ratios and arrangements providing for rotational speed of the fan assembly corresponding to one or more ranges of cruise altitude and/or cruise speed provided above.

Various configurations of the gear assembly provided herein may allow for gear ratios of up to 10:1. Still various examples of the gear assemblies provided herein may allow for gear ratios within a range of 2.5-4.0. Still yet various examples of the gear assemblies provided herein allow for gear ratios within a range of 4.1-10.0. Other examples can have a gear ratio within a range of 3.0-4.0. FIGS. **9**, **12**, **13**, and **16** also provide the gear ratio of several exemplary engines.

Various exemplary gear assemblies are shown and described herein, which can also be referred to as a gearbox. These gear assemblies may be utilized with any of the exemplary engines and/or any other suitable engine for which such gear assemblies may be desirable. In such a manner, it will be appreciated that the gear assemblies disclosed herein may generally be operable with an engine having a rotating element with a plurality of rotor blades and a turbomachinery having a turbine and a shaft rotatable with the turbine. With such an engine, the rotating element (e.g., fan assembly **104**) may be driven by the shaft (e.g., low-speed shaft) of the turbomachinery through the gear assembly.

Although the exemplary gear assemblies shown are mounted at a forward location (e.g., forward from the combustor and/or the low-pressure compressor), in other examples, the gear assemblies described herein can be mounted at an aft location (e.g., aft of the combustor and/or the low-pressure turbine).

Portions of a lubricant system **1700** are depicted schematically in FIG. **21**. The lubrication system **1700** can be a component of the turbomachinery engines disclosed herein (e.g., the engines **100**, **200**, **300**, and **400**) and/or can be coupled to the various gearboxes disclosed herein. For example, FIG. **1** schematically illustrates the lubricant system coupled to the turbofan engine **100** and the gear assembly **102**. FIG. **21** illustrates a series of lubricant conduits **1703** can interconnect multiple elements of the lubricant system **1700** and/or engine components, thereby providing for provision or circulation of the lubricant throughout the lubricant system and any engine components coupled thereto (e.g., a gearbox, bearing compartments, etc.).

It should be understood that the organization of the lubricant system **1700** as shown is by way of example only to illustrate an exemplary system for a turbomachinery engine for circulating lubricant for purposes such as lubrication or heat transfer. Any organization for the lubricant system **1700** is contemplated, with or without the elements as shown, and/or including additional elements interconnected by any necessary conduit system.

Referring still to FIG. **21**, the lubricant system **1700** includes a lubricant reservoir **1702** configured to store a coolant or lubricant, including organic or mineral oils, synthetic oils, or fuel, or mixtures or combinations thereof. A supply line **1704** and a scavenge line **1706** are fluidly coupled to the reservoir **1702** and collectively form a lubricant circuit to which the reservoir **1702** and component **1710** (e.g., a gearbox) can be fluidly coupled. The component **1710** can be supplied with lubrication by way of a fluid coupling with the supply line **1704** and can return the supplied lubricant to the reservoir **1702** by fluidly coupling to the scavenge line **1706**. More specifically, a component supply line **1711** can be fluidly coupled between the supply line **1704** and the component **1710**. It is further contemplated that multiple types of lubricant can be provided in other lines not explicitly shown but are nonetheless included in the lubricant system **1700**.

Optionally, at least one heat exchanger **1705** can be included in the lubricant system **1700**. The heat exchanger **1705** can include a fuel/lubricant (fuel-to-lubricant) heat exchanger, an oil/lubricant heat exchanger, an air-cooled oil cooler, and/or other means for exchanging heat. For example, a fuel/lubricant heat exchanger can be used to heat or cool engine fuel with lubricant passing through the heat exchanger. In another example, a lubricant/oil heat exchanger can be used to heat or cool additional lubricants passing within the turbomachinery engine, fluidly separate from the lubricant passing along the lubricant system **1700**. Such a lubricant/oil heat exchanger can also include a servo/lubricant heat exchanger. Optionally, a second heat exchanger (not shown) can be provided along the exterior of the core engine, downstream of the outlet guide vane assembly. The second heat exchanger can be an air/lubricant heat exchanger, for example, adapted to convectively cool lubricant in the lubricant system **1700** utilizing the airflow passing through an outlet guide vane assembly of the turbomachinery engine.

A pump **1708** can be provided in the lubricant system **1700** to aid in recirculating lubricant from the reservoir **1702** to the component **1710** via the supply line **1704**. For example, the pump **1708** can be driven by a rotating component of the turbomachinery engine, such as a high-pressure shaft or a low-pressure shaft of a turbomachinery engine.

Lubricant can be recovered from the component **1710** by way of the scavenge line **1706** and returned to the reservoir **1702**. In the illustrated example, the pump **1708** is illustrated along the supply line **1704** downstream of the reservoir **1702**. The pump **1708** can be located in any suitable position within the lubricant system **1700**, including along the scavenge line **1706** upstream of the reservoir **1702**. In addition, while not shown, multiple pumps can be provided in the lubricant system **1700**.

In some examples, a bypass line **1712** can be fluidly coupled to the supply line **1704** and scavenge line **1706** in a manner that bypasses the component **1710**. In such examples, a bypass valve **1715** is fluidly coupled to the supply line **1704**, component supply line **1711**, and bypass line **1712**. The bypass valve **1715** is configured to control a flow of lubricant through at least one of the component supply line **1711** or the bypass line **1712**. The bypass valve **1715** can include any suitable valve including, but not limited to, a differential thermal valve, rotary valve, flow control valve, and/or pressure safety valve. In some examples, a plurality of bypass valves can be provided.

During operation, a supply flow **1720** can move from the reservoir **1702**, through the supply line **1704**, and to the bypass valve **1715**. A component input flow **1722** can move from the bypass valve **1715** through the component supply line **1711** to an inlet of the component **1710**. A scavenge flow **1724** can move lubricant from an outlet of the component **1710** through the scavenge line **1706** and back to the reservoir **1702**. Optionally, a bypass flow **1726** can move from the bypass valve **1715** through the bypass line **1712** and to the scavenge line **1706**. The bypass flow **1726** can mix with the scavenge flow **1724** and define a return flow **1728** moving toward the lubricant reservoir **1702**.

In one example where no bypass flow exists, it is contemplated that the supply flow **1720** can be the same as the component input flow **1722** and that the scavenge flow **1724** can be the same as the return flow **1728**. In another example where the bypass flow **1726** has a nonzero flow rate, the supply flow **1720** can be divided at the bypass valve **1715** into the component input flow **1722** and bypass flow **1726**. It will also be understood that additional components, valves, sensors, or conduit lines can be provided in the lubricant system **1700**, and that the example shown in FIG. **21** is simplified with a single component **1710** for purposes of illustration.

The lubricant system **1700** can further include at least one sensing position at which at least one lubricant parameter can be sensed or detected. The at least one lubricant parameter can include, but is not limited to, a flow rate, a temperature, a pressure, a viscosity, a chemical composition of the lubricant, or the like. In the illustrated example, a first sensing position **1716** is located in the supply line **1704** upstream of the component **1710**, and a second sensing position **1718** is located in the scavenge line **1706** downstream of the component **1710**.

In one example, the bypass valve **1715** can be in the form of a differential thermal valve configured to sense or detect at least one lubricant parameter in the form of a temperature of the lubricant. In such a case, the fluid coupling of the bypass valve **1715** to the first and second sensing positions **1716**, **1718** can provide for bypass valve **1715** sensing or detecting the lubricant temperature at the sensing positions **1716**, **1718** as lubricant flows to or from the bypass valve **1715**. The bypass valve **1715** can be configured to control the component input flow **1722** or the bypass flow **1726** based on the sensed or detected temperature.

It is contemplated that the bypass valve **1715**, supply line **1704**, and bypass line **1712** can at least partially define a closed-loop control system for the component **1710**. As used herein, a "closed-loop control system" will refer to a system having mechanical or electronic components that can automatically regulate, adjust, modify, or control a system variable without manual input or other human interaction. Such closed-loop control systems can include sensing components to sense or detect parameters related to the desired variable to be controlled, and the sensed or detected parameters can be utilized as feedback in a "closed loop" manner to change the system variable and alter the sensed or detected parameters back toward a target state. In the example of the lubricant system **1700**, the bypass valve **1715** (e.g., mechanical or electrical component) can sense a parameter, such as a lubricant parameter (e.g., temperature), and automatically adjust a system variable, e.g., flow rate to either or both of the bypass line **1712** or component **1710**, without need of additional or manual input. In one example, the bypass valve can be automatically adjustable or self-adjustable such as a thermal differential bypass valve. In another example, the bypass valve can be operated or actuated via a separate controller. It will be understood that a closed-loop control system as described herein can incorporate such a self-adjustable bypass valve or a controllable bypass valve.

Turning to FIG. **22**, a portion of the lubricant system **1700** is illustrated supplying lubricant to a particular component **1710** in the form of a gearbox **1750** within a turbomachinery engine. The gearbox can be any of the gearboxes disclosed herein. The gearbox **1750** can include an input shaft **1752**, an output shaft **1754**, and a gear assembly **1755**. In one example, the gear assembly **1755** can be in the form of an epicyclic gear assembly as known in the art having a ring gear, sun gear, and at least one planet/star gear. An outer housing **1756** can at least partially surround the gear assembly **1755** and form a structural support for the gears and bearings therein. Either or both of the input and output shafts **1752**, **1754** can be coupled to the turbomachinery engine. In one example, the input and output shafts **1752**, **1754** can be utilized to decouple the speed of the low-pressure turbine from the low-pressure compressor and/or the fan, which can, for example, improve engine efficiency.

The supply line **1704** can be fluidly coupled to the gearbox **1750**, such as to the gear assembly **1755**, to supply lubricant to gears or bearings to the gearbox **1750** during operation. The scavenge line **1706** can be fluidly coupled to the gearbox **1750**, such as to the gear assembly **1755** or outer housing **1756**, to collect lubricant. The bypass line **1712** can be fluidly coupled to the bypass valve **1715**, supply line **1704**, and scavenge line **1706** as shown. A return line **1714** can also be fluidly coupled to the bypass valve **1715**, such as for directing the return flow **1728** to the lubricant reservoir **1702** for recirculation. While not shown in FIG. **22** for brevity, the lubricant reservoir **1702**, the heat exchanger **1705**, and/or the pump **1708** (FIG. **21**) can also be fluidly coupled to the gearbox **1750**. In this manner, the supply line **1704**, bypass line **1712**, scavenge line **1706**, and return line **1714** can at least partially define a recirculation line **1730** (FIG. **21**) for the lubricant system **1700**.

The supply flow **1720** divides at the bypass line into the component input flow **1722** and the bypass flow **1726**. In the example shown, the bypass valve **1715** is in the form of a differential thermal valve that is fluidly coupled to the first and second sensing positions **1716**, **1718**.

Lubricant flowing proximate the first and second sensing positions **1716**, **1718** provides the respective first and second outputs **1741**, **1742** indicative of the temperature of the lubricant at those sensing positions **1716**, **1718**. It will be understood that the supply line **1704** is thermally coupled to the bypass line **1712** and bypass valve **1715** such that the temperature of the fluid in the supply line **1704** proximate the first sensing position **1716** is approximately the same as fluid in the bypass line **1712** adjacent the bypass valve **1715**. Two values being "approximately the same" as used herein will refer to the two values not differing by more than a predetermined amount, such as by more than 20%, or by more than 5 degrees, in some examples. In this manner, the bypass valve **1715** can sense the lubricant temperature in the supply line **1704** and scavenge line **1706** via the first and second outputs **1741**, **1742**. It can be appreciated that the bypass line **1712** can form a sensing line for the valve **1715** to sense the lubricant parameter, such as temperature, at the first sensing position **1716**.

During operation of the turbomachinery engine, the lubricant temperature can increase within the gearbox **1750**, such as due to heat generation of the gearbox **1750**, and throughout the lubricant system **1700**. In one example, if a lubricant temperature exceeds a predetermined threshold temperature at either sensing position **1716**, **1718**, the bypass valve **1715** can automatically increase the component input flow **1722**, e.g., from the supply line **1704** to the gearbox **1750**, by decreasing the bypass flow **1726**. Such a predetermined threshold temperature can be any suitable operating temperature for the gearbox **1750**, such as about 300° F. in some examples. Increasing the component input flow **1722** can provide for cooling of the gearbox **1750**, thereby reducing the lubricant temperature sensed in the various lines **1704**, **1706**, **1712**, **1714** as lubricant recirculates through the lubricant system **1700**.

In another example, if a temperature difference between the sensing positions **1716**, **1718** exceeds a predetermined threshold temperature difference, the bypass valve **1715** can automatically increase the component input flow **1722** by decreasing the bypass flow **1726**. Such a predetermined threshold temperature difference can be any suitable operating temperature for the gearbox **1750**, such as about 70° F., or differing by more than 30%, in some examples. In yet another example, if a temperature difference between the sensing positions **1716**, **1718** is below the predetermined threshold temperature difference, the bypass valve **1715** can automatically decrease the component input flow **1722** or increase the bypass flow **1726**. In this manner the lubricant system **1700** can provide for the gearbox **1750** to operate with a constant temperature difference between the supply and scavenge lines **1704**, **1706**.

This written description uses examples to disclose the technology, including the best mode, and also to enable any person skilled in the art to practice the disclosed technology, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosed technology is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Further aspects of the disclosure are provided by the subject matter of the following clauses:

A turbomachinery engine comprising a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine comprises 3-4 rotating stages. Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and the area ratio is within a range of 2.0-5.1. The gearbox includes an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, and the output of the gearbox is coupled to the fan assembly and comprises a second rotational speed.

The turbomachinery engine of the preceding clause, wherein the area ratio of the low-pressure turbine is within a range of 2.0-3.0.

The turbomachinery engine of any preceding clause, wherein the area ratio of the low-pressure turbine is within a range of 2.2-2.7.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly three rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.2-2.91.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly four rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.0-5.1.

The turbomachinery engine of any preceding clause, wherein the fan assembly is a ducted fan assembly disposed radially within a fan case.

The turbomachinery engine of any preceding clause, wherein the fan assembly is an unducted fan assembly.

The turbomachinery engine of any preceding clause, wherein the fan assembly comprises a first fan and a second fan, each comprising a plurality of fan blades, wherein the second fan is disposed aft of the first fan and has a smaller diameter than the first fan, and wherein the turbomachinery engine is a three-stream engine.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine comprises an AN^2 value within a range of 20-70, where A is the annular exit area of the aft-most rotating stage of the low-pressure turbine measured in square inches, N is the rotational speed of the low-pressure turbine measured in revolutions per minute at a redline operating condition, and the product of AN^2 is divided by 10^9.

The turbomachinery of any preceding clause, wherein the low-pressure turbine further comprises an area-EGT ratio within a range of 1.05-1.6, wherein the

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{(1/(LPT\,stages-1))}}{(EGT/1000)},$$

where the LPT stages is the number of rotating stages of the low-pressure turbine, and the EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition.

The turbomachinery engine of any preceding clause, wherein an/the exhaust gas temperature of the low-pressure turbine is within a range of 1060-1180 degrees Celsius measured at an/the inlet of the low-pressure turbine at a/the redline operating condition.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.0-3.5.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.5-3.5.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.75-4.1.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.0-4.1.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.5.

Example 18. The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-4.0.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.3-3.3.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.3.

A turbomachinery engine comprising a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine comprising 3-4 rotating stages. Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and the area ratio is within a range of 2.0-5.1. The gearbox including an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, the output of the gearbox is coupled to the fan assembly and has a second rotational speed, and a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.5.

The turbomachinery engine of any preceding clause, wherein the area ratio of the low-pressure turbine is within a range of 2.2-3.2.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly three rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.2-2.6.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly four rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.3-2.7.

The turbomachinery engine of any preceding clause, wherein the fan assembly is a ducted fan assembly.

The turbomachinery engine of any preceding clause, wherein the ducted fan assembly comprises a first ducted fan and a second ducted fan, each comprising a plurality of fan blades, wherein the second ducted fan is disposed aft of the first ducted fan and has a smaller diameter than the first ducted fan, and wherein the turbomachinery engine is a three-stream engine.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine comprises an AN^2 value within a range of 20-70, where A the annular exit area of the aft-most rotating stage of the low-pressure turbine measured in square inches, N is the rotational speed of the low-pressure turbine measured in revolutions per minute at a redline operating condition, and a product of AN^2 is divided by 10^9.

The turbomachinery of any preceding clause, wherein the low-pressure turbine further comprises an area-EGT ratio within a range of 1.05-1.6, wherein the

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{(1/(LPT\,stages-1))}}{(EGT/1000)},$$

where the LPT stages is a number of rotating stages of the low-pressure turbine, and the EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition.

The turbomachinery engine of any preceding clause, wherein an/the exhaust gas temperature of the low-pressure turbine is within a range of 1060-1180 degrees Celsius measured at an/the inlet of the low-pressure turbine at a/the redline operating condition.

The turbomachinery engine of any preceding clause, wherein the fan assembly comprises 16-22 fan blades, and wherein the turbomachinery engine further comprises a low-pressure compressor comprising 1-8 stages, a high-pressure compressor comprising 8-11 stages, and a high-pressure turbine comprising 1-2 stages.

The turbomachinery engine of any preceding clause, wherein: the fan assembly comprises 18-20 fan blades; the low-pressure compressor comprises 3-4 stages; the high-pressure compressor comprises 8-9 stages; and the high-pressure turbine comprises 2 stages.

A turbomachinery engine comprising a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly includes a plurality of fan blades. The low-pressure turbine comprises 3-4 rotating stages. Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and the area ratio is within a range of 2.0-5.1. The gearbox includes an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, the output of the gearbox is coupled to the fan assembly and has a second rotational speed, and a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.3-3.3.

The turbomachinery engine of any preceding clause, wherein the gear ratio is within a range of 3.0-3.3.

The turbomachinery engine of any preceding clause, wherein the area ratio of the low-pressure turbine is within a range of 2.0-2.6.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly four rotating stages.

The turbomachinery engine of any example herein, and particularly any one of examples 32-34, wherein the low-pressure turbine includes exactly three rotating stages.

The turbomachinery engine of any preceding clause, wherein the fan assembly is an unducted fan assembly.

The turbomachinery engine of any preceding clause, further comprising a ducted fan assembly disposed aft of the unducted fan assembly, and wherein the turbomachinery engine is a three-stream engine.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine comprises an AN^2 value within a range of 20-70, where A is the annular exit area of the aft-most rotating stage of the low-pressure turbine measured in square inches, N is the rotational speed of the low-pressure turbine measured in revolutions per minute at a redline operating condition, and a product of AN^2 is divided by 10^9.

The turbomachinery of any preceding clause, wherein the low-pressure turbine further comprises an area-EGT ratio within a range of 1.05-1.6, wherein the

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{(1/(LPT\,stages-1))}}{(EGT/1000)},$$

turbine, and the EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition.

The turbomachinery engine of any preceding clause, wherein an/the exhaust gas temperature of the low-pressure turbine is within a range of 1060-1180 degrees Celsius measured at an/the inlet of the low-pressure turbine at a/the redline operating condition.

The turbomachinery engine of any preceding clause, wherein the fan assembly comprises 8-22 fan blades. The turbomachinery engine further comprises a low-pressure compressor comprising 1-5 stages, a high-pressure compressor comprising 7-11 stages, a high-pressure turbine comprising 1-2 stages.

The turbomachinery engine of any preceding clause, wherein: the fan assembly comprises 12-18 fan blades; the low-pressure compressor comprises 3-4 stages; the high-pressure compressor comprises 8-10 stages; and the high-pressure turbine comprises 2 stages.

A turbomachinery engine comprising a fan assembly, a low-pressure turbine, and a gearbox. The fan assembly including a plurality of fan blades. The low-pressure turbine comprises 3-4 rotating stages and an area-EGT ratio within a range of 1.05-1.6. The

$$\text{area-}EGT\text{ ratio} = \frac{(\text{area ratio})^{(1/(LPT\ stages-1))}}{(EGT/1000)}.$$

Each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio is the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine. The LPT stages is the number of rotating stages of the low-pressure turbine. The EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition. The gearbox including an input and an output. The input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, and the output of the gearbox is coupled to the fan assembly and has a second rotational speed.

The turbomachinery engine of any preceding clause, wherein the area-EGT ratio is within a range of 1.05-1.3.

The turbomachinery engine of any preceding clause, wherein the area-EGT ratio is within a range of 1.31-1.53.

The turbomachinery engine of any preceding clause, wherein the area-EGT ratio is within a range of 1.2-1.36.

The turbomachinery engine of any preceding clause, wherein the area ratio of the low-pressure turbine is within a range of 2.0-3.5.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly three rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.0-2.91.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine includes exactly four rotating stages, and wherein the area ratio of the low-pressure turbine is within a range of 2.0-3.2.

The turbomachinery engine of any preceding clause, wherein the fan assembly is a ducted fan assembly disposed radially within a fan case.

The turbomachinery engine of any preceding clause, wherein the fan assembly is an unducted fan assembly.

The turbomachinery engine of any preceding clause, wherein the fan assembly comprises a first fan and a second fan, each comprising a plurality of fan blades, wherein the second fan is disposed aft of the first fan and has a smaller diameter than the first fan, and wherein the turbomachinery engine is a three-stream engine.

The turbomachinery engine of any preceding clause, wherein the low-pressure turbine comprises an AN^2 value within a range of 25-70, where A is the annular exit area of the aft-most rotating stage of the low-pressure turbine measured in square inches, N is the rotational speed of the low-pressure turbine measured in revolutions per minute at a redline operating condition, and a product of AN^2 is divided by 10^9.

The turbomachinery engine of any preceding clause, wherein the exhaust gas temperature of the low-pressure turbine is within a range of 1060-1180 degrees Celsius measured at the inlet of the low-pressure turbine at the redline operating condition.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.0-4.0.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.5-3.5.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.3.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.75-3.5.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-4.0.

The turbomachinery engine of any preceding clause, wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 2.25-3.25.

A turbine for an aircraft engine comprising 3-4 rotating stages and an area ratio within a range of 2.0-5.1. Each rotating stage comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio equals the annular exit area of an aft-most rotating stage divided by the annular exit area of a forward-most rotating stage.

The turbine of any preceding clause, wherein the turbine is a low-pressure turbine disposed aft of a high-pressure turbine.

A turbine for an aircraft engine comprising 3-4 rotating stages and an area-EGT ratio within a range of 1.05-1.6. The area-EGT ratio=(area ratio)$^{(1/(stages-1))}$/(EGT/1000). Each rotating stage comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius. The area ratio is the annular exit area of an aft-most rotating stage divided by the annular exit area of a forward-most rotating stage, wherein the stages is the number of rotating stages. The EGT is an exhaust gas temperature measured in degrees Celsius at an inlet of the turbine at a redline operating condition.

The turbine of any preceding clause, wherein the turbine is a low-pressure turbine disposed aft of a high-pressure turbine.

The invention claimed is:

1. A turbomachinery engine comprising:

a fan assembly comprising a plurality of fan blades, wherein the fan assembly comprises a diameter within a range of 78-84 inches;

a low-pressure compressor comprising exactly three stages;

a high-pressure compressor comprising exactly nine-8-11 stages;

a combustor;

a high-pressure turbine comprising exactly two stages;

a low-pressure turbine comprising 3-4 rotating stages, wherein each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius, wherein the low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and wherein the area ratio is within a range of 2.0-5.1; and

a gearbox including an input and an output, wherein the input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, wherein the output of the gearbox is coupled to the fan assembly and has a second rotational speed, and wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.3.

2. The turbomachinery engine of claim **1**, wherein the low-pressure turbine further comprises an area-EGT ratio within a range of 1.2-1.3,

wherein the

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{\left(\frac{1}{LPT\ stages-1}\right)}}{\left(\frac{EGT}{1000}\right)},$$

wherein the LPT stages is a number of rotating stages of the low-pressure turbine, and

wherein the EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition.

3. The turbomachinery engine of claim **1**, wherein the fan assembly comprises exactly 20 fan blades.

4. The turbomachinery engine of claim **1**, wherein the high-pressure compressor comprises exactly nine stages.

5. The turbomachinery engine of claim **1**, wherein the low-pressure turbine comprises exactly four rotating stages.

6. A turbomachinery engine comprising:

a fan assembly comprising exactly 20 fan blades, wherein the fan assembly comprises a diameter within a range of 78-84 inches;

a low-pressure compressor comprising exactly three stages;

a high-pressure compressor comprising exactly nine stages;

a combustor;

a high-pressure turbine comprising exactly two stages;

a low-pressure turbine comprising exactly four rotating stages, wherein each rotating stage of the low-pressure turbine comprises an annular exit area defined by a tip radius of a trailing edge of any one blade of the rotating stage and a hub radius of the any one blade of the rotating stage at an axial location aligned with the tip radius, wherein the low-pressure turbine comprises an area ratio equal to the annular exit area of an aft-most rotating stage of the low-pressure turbine divided by the annular exit area of a forward-most rotating stage of the low-pressure turbine, and wherein the area ratio is within a range of 2.55-2.65; and

a gearbox including an input and an output, wherein the input of the gearbox is coupled to the low-pressure turbine and comprises a first rotational speed, wherein the output of the gearbox is coupled to the fan assembly and has a second rotational speed, and wherein a gear ratio of the first rotational speed to the second rotational speed is within a range of 3.0-3.3.

7. The turbomachinery engine of claim **6**, wherein the low-pressure turbine further comprises an area-EGT ratio within a range of 1.2-1.3,

wherein the

$$\text{area-}EGT\text{ ratio} = \frac{(\text{the area ratio})^{\left(\frac{1}{LPT\ stages-1}\right)}}{\left(\frac{EGT}{1000}\right)},$$

wherein the LPT stages is a number of rotating stages of the low-pressure turbine, and

wherein the EGT is an exhaust gas temperature of the low-pressure turbine measured in degrees Celsius at an inlet of the low-pressure turbine at a redline operating condition.

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